

OpenURMA: An Open-Source FPGA Implementation of UB’s Connectionless URMA (Unified Remote Memory Access) Transport

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Datacenter-scale RDMA fabrics that connect N local applications to M remote endpoints pay an $O(N \cdot M)$ per-NIC state cost under standard RoCE/InfiniBand: each application opens a separate Queue Pair to each endpoint, and each QP carries multi-hundred-byte state. Huawei’s UB (Unified Bus) specification defines a different design point along three architectural axes simultaneously: **(i) scalability** — a connectionless transport whose state grows as $O(N+M)$ via a Jetty (per-application endpoint) plus TP Channel (per-remote-host transport) split; **(ii) ordering** — a graded surface (four service modes $\times$ three execution tags $\times$ Fence $\times$ in/out-of-order completion) that lets applications opt into precisely the consistency they need; and **(iii) latency** — a memory-semantic data path (§8.3 Load/Store synchronous access paired with §6.1 TP Bypass) in which CPU `ld/st` instructions reach remote DRAM through an on-chip-bus-attached controller, removing the PCIe round trip that anchors the per-op floor of every PCIe-attached RDMA NIC.

We present *OpenURMA*, an open-source clean-room FPGA implementation of UB’s transaction and transport layers, built as 38 `.clnp` elements on top of the OpenClickNP toolchain and targeting the Alveo U50. We implement the full UB §7.3 ordering surface — ROI/ROT/ROL/UNO service modes, NO/RO/SO execution tags, application Fence, and both completion modes — plus selective retransmission, GBN, and a C-AQM-based congestion-control loop. To anchor the comparison we build a parallel RoCEv2 RC reference (*OpenRoCE*, 21 elements) on identical infrastructure.

We evaluate the design through three independent paths on *both* stacks: sixteen SW-emu correctness tests on OpenURMA (all passing, including a multi-flit Write payload test that exercises 8/32/64/128/256 B writes through Eth_Encap streaming, Eth_Decap reassembly, and HBM_Write payload byte stream; a ROL fused-ack test that validates §7.3.3.4’s TPack/TAACK fusion; and a full-atomic-opcode test that covers Swap/Store/Load/FAA/FSUB/FAND/FOR/FXOR per §7.4.2.3) plus eight matched-shape tests on OpenRoCE (all passing); a SystemC microbench whose seven OpenURMA scenarios (per-stage latency, per-mode TX, throughput, Fence cost, SO blocking depth, cross-INI HOL bypass, payload sweep) and five matched OpenRoCE scenarios run on the same harness; and a Vivado out-of-context synth+place&route sweep over every element of both stacks. All 38 OpenURMA elements and all 21 OpenRoCE elements close 322 MHz post-route. At ($N=1024$, $M=1024$) endpoints OpenURMA holds **4,855 $\times$ less per-connection NIC state** and **20.8 $\times$ less host-side memory** than OpenRoCE, while delivering **2.80 $\times$ higher sustained throughput** (150.4 vs 53.6 WR/ μ s, paced microbench) at the cost of 46.6 ns more first-flit latency (24 vs 9 cycles cold) and 2.63 $\times$ more LUT area. Per-stage breakdown localises the area budget: Pillar 2 (ordering / completion) costs 13 KLUT (10.6% of OpenURMA’s total); transport reliable-delivery costs 37 KLUT (30.2%); header parse/build is 18 KLUT (15.0%); data path is 19 KLUT (15.3%), inclusive of the multi-flit Write payload byte-stream machinery.

As a side contribution we extend OpenClickNP’s `.clnp` DSL with two per-element pragmas (`.timing { ii = N; }` and `.hls_pragma { ... }`) that make HLS guidance a first-class language feature.

1. Introduction

For modern high-fanout RDMA workloads — AI training collectives, HPC all-to-all, parameter-server fan-in/fan-out — the binding NIC resource is no longer link bandwidth but the *cardinality* of active connections each NIC has to keep state for. A 1024-application AI training job whose participants exchange tensors with every other peer holds 1024 “sender-side” resources $\times$ 1024 “receiver-side” targets. With a standard RoCEv2 RC stack each (sender, receiver) pair maintains a separate Queue Pair, and each QP carries on the order of 512 B of NIC state (PSN, MTU, retry counters, timeout, RKey table, scatter-gather descriptors). The total NIC state is then $1024 \times 1024 \times 512 \text{ B} = 537 \text{ MB}$ — well past what fits in a typical NIC’s on-chip BRAM, and

well into the regime where the NIC has to spill to host memory across PCIe, paying both PCIe latency and host DRAM bandwidth on every DMA. This is the QP scaling problem [19, 14, 16]; it has generated several lines of mitigation work [27, 42, 37], including hardware connection-cache schemes, software multiplexing, and connectionless transports built atop UDP.

Huawei’s Unified Bus (UB) specification [11] attacks the problem at the transport-layer design itself. UB’s user-facing verb surface is called URMA (Unified Remote Memory Access), the same name we adopt for the FPGA implementation we present here. *Three architectural pillars* are load-bearing for our argument:

1. **Scalability — Transaction / Transport split.** A *Jetty* is the per-application endpoint (analogous to a QP, but

connectionless); a *TP Channel* is the per-remote-host transport that carries reliable delivery state. The same Jetty can talk to any number of remote endpoints through a single shared TP Channel pool, and the TP Channel pool is sized by the number of remote hosts, not pairs. State scales as $O(N+M)$ rather than $O(N \cdot M)$.

2. **Ordering — graded surface.** UB §7.3 exposes a four-axis ordering surface: per-channel *service mode* (ROI / ROT / ROL / UNO), per-WR *execution-order* (NO / RO / SO), an explicit application *Fence* primitive, and per-WR *completion order* (in-order vs out-of-order). Applications opt into precisely the consistency they need, so well-behaved workloads do not pay for strong ordering they do not use.
3. **Latency — memory-semantic data path.** UB §8.3 defines *Load/Store synchronous access*: the CPU issues bare ISA `ld/st` instructions, the on-chip-bus carries them to a UB Controller (on the same die as the CPU, *not* behind PCIe), and the controller lowers each one into a single-shot UB transaction tagged with `NLP = TP Bypass` (§6.1). No WQE, no doorbell, no CQE poll. This removes the four PCIe traversals (doorbell MMIO + DMA-WQE-fetch + DMA-CQE-write + CPU CQE-poll-miss) that anchor the per-op latency floor of every PCIe-attached RDMA NIC, and replaces them with on-chip-bus crossings whose cost is $10\times$ smaller.

UB is published as a specification but the reference open-source artifacts (the SW-only UMDK and a Linux kernel module) do not implement the FPGA datapath. Huawei has separately disclosed the *Ascend 950* NPU series [12] as the first publicly documented silicon implementing UB — with the §8.3 memory-semantic path (“UB Memory”: Load/Store + AtomicStore/Load/Swap/CAS) and the §8.4 URMA-async path (Jetty-driven Write/Read/Send/Atomic) shipping as separate on-die blocks, the same split this paper preserves in RTL — but the silicon is closed and not available for FPGA-comparable experiments. This paper presents *OpenURMA*, an open-source FPGA implementation that exercises all three pillars in synthesisable RTL and quantifies their cost.

Contributions.

- A 38-element clean-room implementation of UB’s transaction and transport layers (3.4 KLOC of `.clnp` for the in-topology elements; 3.7 KLOC including four out-of-topology aux elements — TPSACK, Switch_CAQM, and the two AXI-MM HBM variants used by the FPGA-link backend), covering the full §7.3 ordering surface plus selective and Go-Back-N retransmission, a C-AQM-based switch model and Initiator-side window controller, the full UB Atomic opcode set (CAS / Swap / FAA / FSUB / FAND / FOR / FXOR / Store / Load), an RTO timer with exponential backoff, the multi-path TPG element of UB §6.3.2, and a multi-flit Write payload path that streams 8–256 B writes through `Eth_Encap`, `Eth_Decap`, and the `HBM_Write` byte stream. All 38 elements close 322 MHz on Vivado out-of-context P&R targeting Alveo U50.
- A parallel 21-element RoCEv2 RC reference (*Open-*

RoCE) on the same OpenClickNP toolchain and target part, with matched test coverage: the OpenRoCE baseline ships 8 SW-emu tests (roundtrip, throughput, mem-data-integrity, dcqcn-end-to-end, HOL behaviour, multi-QP parallelism, atomic CAS+FAA, tx-wire) that mirror the architecturally-comparable subset of OpenURMA’s 16 tests, plus a 5-scenario SystemC microbench (tx_latency, throughput, per_element, payload, hol_blocking) so each protocol’s numbers come from the same harness.

- A three-path evaluation on *both* stacks: (i) 16 OpenURMA + 8 OpenRoCE SW-emu correctness tests (all 24 passing; OpenURMA tests exercise the multi-flit Write payload path, the ROL fused-ack semantics of §7.3.3.4, and the full §7.4.2.3 atomic opcode surface); (ii) a 7-scenario SystemC microbench on OpenURMA (per-stage latency, per-mode TX, throughput, Fence cost, SO blocking depth, cross-INI HOL bypass, payload sweep) plus the matched 5-scenario microbench on OpenRoCE; (iii) a Vivado out-of-context place-and-route sweep over every element of both stacks with per-category area decomposition.
- **Pillar 1 — scalability.** At $N=M=1024$, OpenURMA holds **4,855 $\times$ less NIC state** and delivers **2.80 $\times$ higher sustained throughput** than OpenRoCE, at $2.63\times$ more LUT area and 46.6 ns more first-flit latency.
- **Pillar 2 — graded ordering.** The four-axis §7.3 surface costs only 13 KLUT — 10.6% of OpenURMA’s total — for a feature set that has no equivalent in RoCE, and adds *zero* extra pipeline cycles on paths that do not request ordering.
- **Pillar 3 — memory-semantic latency.** A new `UB_LoadStore_Engine` element and the `examples/openurma_loadstore/topology` variant wire `doorbell` $\rightarrow$ `ldst` $\rightarrow$ `btah_b` $\rightarrow$ `nth_b` $\rightarrow$ `ethenc` (5 stages, vs the URMA-async path’s 10). At 322 MHz the first wire flit emerges at **8 cycles (≈ 25 ns)** — 17 cycles below the URMA-async cold path on identical infrastructure. The saving is structural: TP Bypass removes `Cwnd` + `Retrans` + `RTPH` state from the NIC budget for workloads that opt in.
- **A two-node end-to-end simulator** (`eval/twonode/`, SystemC) that links the three NIC stacks as static libraries and models the surrounding subsystems — CPU-side work-request construction, doorbell PIO, NIC-initiated DMA WQE fetch, NIC pipeline, EtherLink, remote DRAM, NIC-initiated DMA CQE write, CPU-side CQE poll, and a write-back / write-through / uncacheable two-level cache hierarchy — each as an explicit SystemC component on the critical path. The simulator covers all six verb categories (Load/Store, Read, Write, Send/Recv, Atomic) and reports end-to-end CPU-to-DRAM-to-CPU latency with full bidirectional accounting (initiator-side host $\leftrightarrow$ NIC, target-side NIC $\leftrightarrow$ DRAM, target DRAM, peer response, return wire, and CPU completion all modeled explicitly). UB Load/Store reaches **≈ 500 ns end-to-end** on a READ-class verb at 100 ns link delay, **4.37 $\times$ lower than a matched RoCEv2 RC DMA baseline (2186 ns)**; the gap is dominated by the four PCIe traversals (initiator DMA

WQE fetch + initiator DMA CQE write + target DMA-read of host memory + initiator DMA-write of returned payload to host memory) that RoCE pays on every READ operation. The full sweep is 388 configurations (eval/twonode/results/sweep.csv).

- Two upstreamed extensions to the OpenClickNP .clnp DSL — per-element .timing { ii = N; } and .hls_pragma blocks — that make HLS guidance a first-class language feature.

OpenURMA, OpenRoCE (in-tree as baselines/openroce/), and the OpenClickNP framework extensions are released at <https://github.com/bojieli/OpenURMA> and <https://github.com/bojieli/OpenClickNP>.

2. Background

2.1 The QP scaling problem

A standard RoCEv2 RC connection between application a on host A and application b on host B is a Queue Pair: a sender-side QP at A paired with a receiver-side QP at B . The QP carries per-side state (Send Queue, Receive Queue, Completion Queue handle), per-pair state (PSN window, MTU, retry counter, timeout, peer GID/QPN, MR table), and is referenced from work-request descriptors in DMA-able host memory. Production NICs implement the QP context as a ~ 256 -512 B record in on-chip SRAM [33, 15]; on-chip space is finite, so beyond a few thousand active QPs the NIC spills to host memory and pays a PCIe-round-trip per RDMA op.

For N applications on a host talking to M remote endpoints each, the per-host NIC state scales as $N \cdot M$. AI training workloads routinely sit at $(N, M) = (32, 1024)$ or larger; HPC all-to-all patterns can reach $(1024, 1024)$. Connection-cache schemes (SRNIC [42], FaSST [14]) and shared-state designs (IRMA [37]) reduce the per-connection footprint but do not change the asymptotic scaling.

2.2 UB’s solution: Jetty + TP Channel

UB introduces two distinct abstractions.

Jetty is an application-level RDMA endpoint, identified by a 24-bit Jetty ID. It owns Send / Receive / Completion / Read-Response queues. It is *connectionless*: there is no peer-Jetty handle in its state. Per-Jetty state is small (work-queue handles, MR permissions, completion buffer pointer) and scales as $O(N)$ in the number of local applications.

TP (Transport) Channel is the per-remote-host reliable-transport state: PSN window, retransmit queue, in-flight bitmap, RTO timer, congestion-window register. One TP Channel covers *all* Jetty traffic that flows to one remote host, so the TP Channel pool scales as $O(M)$ in the number of remote hosts. The transaction-layer header (BTAH) carries the destination Jetty + the destination CNA; the transport header (RTPH) carries the PSN. The Initiator side multiplexes Jetty traffic onto TP Channels by destination host; the Target side demultiplexes by Jetty ID after PSN reordering.

The result is total NIC state $O(N+M)$ rather than $O(N \cdot M)$. Concretely, at $(N, M) = (1024, 1024)$ OpenURMA sits at ≈ 110 KB of NIC state vs OpenRoCE’s ≈ 537 MB, a $4,855\times$ reduction.

2.3 Graded ordering surface

UB §7.3 exposes ordering as a 4-axis design space, not a single knob:

- **Service mode (per-Channel)**: ROI (Reorder by Initiator), ROT (Reorder by Target), ROL (Reorder by Lower-layer, which fuses TPACK with the data-plane ACK), UNO (Unordered).
- **Execution order (per-WR)**: NO (no ordering), RO (Relax-order), SO (Strict-order). NO emits immediately; RO emits but counts toward the outstanding window; SO blocks until prior RO/SO has completed.
- **Fence**: an explicit application-level barrier that serialises all subsequent WRs against all prior RO/SO completions.
- **Completion order (per-WR)**: in-order (ODR [2] = 1, where ODR is the per-WR Order field) holds JFC (Jetty Function Completion queue) entries until all prior INI_TASSNs (per-Initiator Transaction Sequence Numbers) from the same Initiator have completed; out-of-order (ODR [2] = 0) emits arrival-order.

The four-axis surface lets a workload that doesn’t need ordering (a pure-fanout broadcast under UNO + NO) skip every gating stage on the NIC, while a workload that does need ordering (a barrier-after- collective under ROI + SO + Fence) gets it without paying global serialisation. RoCE’s RC, by contrast, enforces strict in-order delivery on every operation regardless of need.

2.4 OpenClickNP

OpenClickNP [21] is a clean-room reimplement of the ClickNP element model [23] on top of Alveo U50. The unit of design is a .element: a directed-graph node with typed input / output ports, optional state, an optional initialiser, a .handler block (the per-cycle behaviour), and an optional .signal block (the host-RPC control surface). Elements compose through a topology.clnp file that wires ports together.

The toolchain compiles .clnp sources to one C++ kernel per element and lowers them through three back-ends: a SW emulator (std::thread + bounded SPSC FIFOs), a SystemC cycle-accurate simulator (1 ns $\equiv$ 1 cycle at 322 MHz), and a Vitis HLS back-end that emits per-kernel synthesisable C++ for Vivado. Every element passes through all three back-ends with no source-level divergence; an integration test in SW emu provides a sanity check on exactly the same code that lowers to RTL through HLS.

Framework extensions for OpenURMA. Six OpenURMA elements (atom, comp_reord, ord_ini, ord_tgt, jsched, mr_tab) needed pipeline-initialisation-interval guidance to close 322 MHz Vivado P&R; the OpenClickNP back-end previously hard-coded #pragma HLS PIPELINE II=1. We extended the .clnp grammar with two pragma blocks:

```
.timing { ii = 2; }
.hls_pragma {
    "ARRAY_PARTITION variable=_state.hbm
    cyclic factor=8 dim=1";
}
```

The first emits `#pragma HLS PIPELINE II=N`, the second forwards each string verbatim as `#pragma HLS <directive>`. Both are upstream in `OpenClickNP`. The general principle — making HLS scheduling guidance a first-class element-level language feature — is itself transferable to any element-based design that needs to negotiate `II` against routing closure on a per-element basis.

3. Design

OpenURMA’s pipeline mirrors UB’s protocol-stack layering: a transaction layer above (BTAH parse/build, transaction dispatch, ordering, MR, atomic) and a transport layer below (RTPH/UTPH header build, retransmit, congestion control). The two layers compose through clearly typed flit boundaries. Figure 1 sketches the topology.

3.1 Element decomposition

The 38 OpenURMA elements break down into six architectural categories (the same categorisation drives the per-category area table in §9). Ten are header parsers/builders shared across both directions (`Eth_Encap/Decap` and the matched parse/build pairs for `NTH`, `RTPH`, `UTPH`, `BTAH`). Nine implement the transport-layer reliable-delivery machinery (`TP_Channel_TX/RX`, `Retrans_Buffer`, `PSN_Reorder`, `TPACK_Gen`, `TAACK_Gen`, `Cong_Window`, `Cong_Echo`, `TPG_Group`). Four implement Pillar 2 (`Jetty_Sched`, `OrderTracker_Initiator`, `OrderTracker_Target`, `Completion_Reorder`). Six are data-path elements (`HBM_Read`, `HBM_Write`, `Atomic`, `Jetty_Recv`, `Txn_Dispatch`, `Completion_Gen`). Three are state tables (`MR_Table`, `Jetty_Table`, `TP_Table`). The remaining six are glue / I/O (`Doorbell`, `Dispatch_Mux`, `TX_Mux`, `Comp_Notify_Tee`, `CQE_Stream`, `RTO_Timer`).

A few decompositions deserve note. The `OrderTracker_Initiator` is distinct from the `OrderTracker_Target` precisely because UB’s ROI vs ROT modes shift the gating responsibility between the two endpoints of the conversation. `Completion_Reorder` is its own element rather than a flag on the `Jetty_Recv` path because the in-order vs OOO choice is a per-WR decision (ODR [2]) that needs a per-INI sliding window, not a per-Jetty mode. `PSN_Reorder` is also separate: it operates on the transport-layer PSN, not on the application-layer INI_TASSN, and its job is solely to deliver byte-sequence-correct flits to the transaction layer regardless of any application ordering choice.

3.2 Topology specifics

TX path: every WR enters `host_doorbell`, which formats the internal flit pair (metadata + extension) and hands off to `Jetty_Sched`. `Jetty_Sched` consults the per-Jetty Fence state and the per-channel outstanding-RO/SO window before admitting the WR

to `OrderTracker_Initiator`; on Fence-active Jetties the WR is queued until prior RO/SO completes. `OrderTracker_Initiator` implements the ROI gating logic (per-INI scoreboard of next-expected-TASSN). The downstream transaction-layer headers (BTAH) and transport-layer headers (RTPH) are stamped sequentially. `Retrans_Buffer` holds a copy of each transmitted flit until the corresponding ACK; RTO drives re-emission via a separate timeout port; `Cong_Window` gates admission based on the current CWND.

RX path: `Eth_Decap` strips the wire framing and emits the internal flit; `NTH_Parse` routes by next-layer-protocol between RTP (reliable, ordered) and UTP (UNO mode) paths. `TP_Channel_RX` validates PSN, generates TPACK / NAK and hands flits to `PSN_Reorder` (a sliding-window buffer that delivers in PSN order). `BTAH_Parse` routes by transaction type; `OrderTracker_Target` implements ROT gating; `MR_Table` validates token IDs and rewrites VA → HBM offset; `Txn_Dispatch` branches by opcode to the right data-path element. `Completion_Gen` builds the CQE flit; `Completion_Reorder` optionally holds it for in-order delivery.

3.3 Ordering machinery in detail

The most subtle elements are the four ordering modules. We highlight two design choices.

OrderTracker_Initiator (ROI). Maintains a per-Initiator (`rc_id & 0xF`) round-robin queue of size 32 entries per INI across 16 INI slots. NO bypasses. RO emits and bumps `outstanding_ro_so`. SO is admitted only if `outstanding_ro_so == 0`; otherwise it is queued. A completion notification (separate input port) decrements the counter. A round-robin scan emits one queued SO per cycle when ready. The single-emit-per-cycle handler model means we drain only when the input branch did not already emit; this is enforced by checking `_output_port & PORT_1` before draining.

Completion_Reorder. A per-INI ring buffer of 8 slots indexed by $(ts - next_tassn) \bmod WIN$. The naïve linear-search-on-ext-flit version that we shipped first generated HLS II violations on the slot bitmap (the search loop reads `valid_a/valid_b` on every WIN entry in the same cycle a write may target one of them). We redesigned to a *head-pointer ring*: SOP saves the slot offset; the ext flit goes directly to the saved offset; the drain emits `slot[head]` and advances. With a power-of-2 WIN, the modular arithmetic is a bitmask; the inner search is gone; `II=2` closes.

3.4 Retransmission and congestion control

`Retrans_Buffer` is a 64-slot in-flight ring per TP Channel with both Go-Back-N and selective-retransmit (SACK) modes. The selective-retransmit path uses the TPSACK response opcode plus a per-PSN bitmap in the TPACK header (UB-Base-Spec Annex B) [11]. The retransmit decision is driven by either a TAACK that explicitly NAKs PSNs

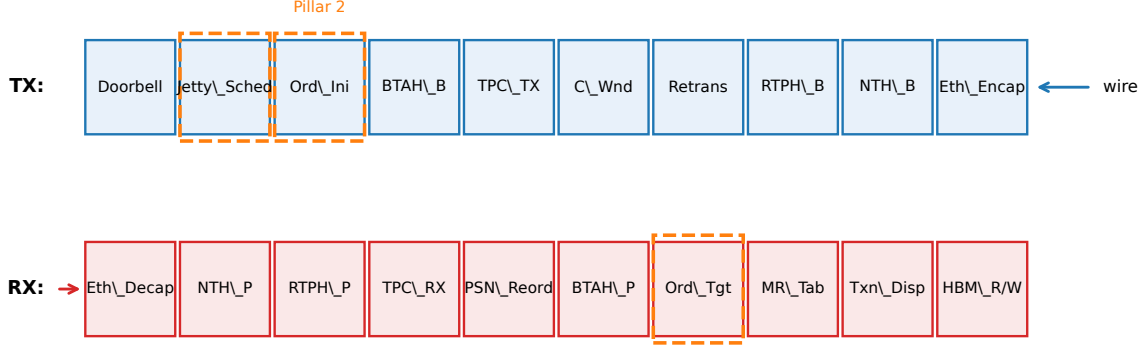


Figure 1: OpenURMA pipeline topology, 38 elements organised by layer. Dashed orange boxes mark Pillar 2 ordering elements. Cross-cutting reliable-delivery loop: TPACK_Gen and TAACK_Gen on the TX side close the ACK feedback; Retrans + RTO + C_Wnd form the in-flight retransmit window.

(selective) or the RTO element firing its single-port timer (Go-Back-N). Each slot today captures one (meta, ext) flit pair, which is sufficient for control-plane WRs and for ≤ 8 B Writes; *multi-flit retransmission* (extending the slot to a payload-flit list so a dropped Eth_Encap-streamed Write can be replayed in full) is not yet implemented in the retrans buffer — the multi-flit Write data path (§6) currently runs only on the loss-free path. Adding payload-flit storage to each slot is flagged as follow-on work in §10.

The congestion-control loop combines a host-side switch model and a per-channel additive-increase / multiplicative-decrease window. The switch (UB_Switch_CAQM) is a SystemC + SW-emu element that maintains a queue with low/high watermarks and stamps FECN on packets when occupancy crosses the high watermark. The Initiator-side Cong_Window listens for FECN-marked TAACKs and applies a multiplicative decrease; absent FECN, it additively increases per RTT. We verify the closed loop in `tests/swemu/test_caqm_endtoend.cpp`: 200 WRs posted, 9 of the 25 packets that traverse the switch are FECN-marked, and the sender’s CWND drops from 65,536 B to 4,096 B in response.

3.5 Pillar 2 in code

The full §7.3 surface is exercised by SW-emu integration tests: `test_roi_ordering` (ROI gates SO behind RO), `test_rot_ordering` (ROT defers SO behind missing TASN), `test_uno` (UNO path bypasses ordering), `test_rol_fused_ack` (ROL’s §7.3.3.4 TPACK/TAACK fusion — TPACK_Gen stamps RSPST=TPACK_W_TAACK on ROL acks while TAACK_Gen drops ROL/UNO response flits because the TPACK already carries the fused ack info), `test_fence` (Fence gates Write behind outstanding Read), `test_mixed_modes` (4-WR stream with mixed ROI/ROL/UNO and NO/RO/SO), and `test_completion_order` (ODR[2]=1 reorders, ODR[2]=0 bypasses). The wider 16-test SW-emu suite (§6) covers the rest of the spec surface plus framework internals; all 16 tests pass on the current source tree.

4. Implementation

The 38 OpenURMA elements total 3.4 KLOC of .clnp. The parallel OpenRoCE baseline is 19 .clnp elements (1.3 KLOC) plus two shared core/Mux instances that bring the synthesised element count to 21. Both stacks sit on the same OpenClickNP toolchain (~3.8 KLOC of compiler plus ~2.2 KLOC of runtime). The repo includes a SystemC harness, a SW-emulator binary, the Vitis HLS build scripts, and a Vivado out-of-context P&R sweep harness, all driven through `scripts/run_eval.sh` for one-click reproducibility.

4.1 Timing-closure sweep

Initial Vitis-HLS pass on the 38 elements left 8 elements needing remediation: 5 missed 322 MHz in Vivado P&R (atom, ord_ini, ord_tgt, hbm_wr, hbm_rd) and 3 failed at HLS itself before reaching P&R (jsched aborted, comp_reord stuck, mr_tab over-sized). The mechanical resolution combined the new .timing/ .hls_pragma pragmas with a few targeted algorithmic redesigns:

Element	Before → After (ns WNS)	Action
atom	−1.871 → +0.298	II=4 + ARRAY_PARTITION
comp_reord	HLS-stuck → +0.672	head-pointer ring + II=2
ord_ini	−5.382 → +0.318	II=2
jsched	HLS-aborted → +0.546	II=2
ord_tgt	−2.25 → +0.101	II=2 + drain re-order
hbm_wr	−0.628 → +0.628	ARRAY_PARTITION
mr_tab	failing → +0.729	MAX_MR 256 → 64
hbm_rd	−0.449 → +0.282	uint64_t word-array

The hbm_rd story. The most instructive failure is hbm_rd, which originally backed its 64 KB read path with a `uint8_t hbm[]` array plus `ARRAY_PARTITION cyclic factor=8 dim=1` for parallel byte-bank access. HLS estimated 490 MHz Fmax; Vivado P&R reported −0.449 ns WNS. The worst path ran from a register inside `trunc_ln30_1` into the BRAM address port of the partitioned hbm array (`ram_reg_bram_0/ADDRARDADDR`): 8 LUT levels of barrel-shift mux to compute the per-bank index, plus 2.7 ns of routing delay (76% of the period).

No combination of II or partition factor moved the slack: the wide muxing for the 8 banks bloated routing without unblocking timing.

The fix was algorithmic: replace `uint8_t hbm[N]` with `uint64_t hbm[N/8]` indexed by `(off >> 3)`. Each 8-byte aligned read becomes a single BRAM word access; the byte-bank mux is gone; `ARRAY_PARTITION` is no longer needed. WNS closes at +0.282 ns. The same pattern was applied to OpenRoCE’s `RoCE_Mem_Read`.

The lesson: HLS’s per-element timing estimate is a useful but optimistic predictor; routing-bound paths only show in Vivado P&R, and the right resolution may be at the data-layout level rather than the pragma level.

4.2 Test coverage

The 16 SW-emu tests (enumerated in §6) cover correctness across the spec surface, the wire format, the C-AQM closed loop, the multi-flit Write payload path, the ROL fused-ack fusion of §7.3.3.4, and the full §7.4.2.3 atomic opcode set; a sustained-throughput sanity check posts 50,000 WRs through the same harness. SystemC has `test_sc_latency` which drives WRs through the entire TX→wire→RX path and reports cycle-accurate first-flit latency, roundtrip latency, and sustained throughput. The HLS sweep + Vivado P&R sweep are driven by `scripts/synth_hls.sh` and `scripts/vivado_all.sh` respectively.

5. Evaluation Methodology

Because no FPGA is available for in-silicon experiments, we evaluate the design through three independent paths, each answering a different class of question:

- **SW-emulator (functional correctness).** Each `.clnp` element compiles to a C++ kernel that runs in a `std::thread` with bounded SPSC FIFOs between elements. Tests post WRs into the pipeline’s input stream, drain responses, and assert spec-defined behaviour.
- **SystemC (cycle-accurate).** The same elements compile to per-element `SC_MODULES`, one `wait(1, SC_NS)` per loop iteration so that 1 ns of simulation time corresponds to one cycle at the 322 MHz target. We measure cycle counts directly and convert to wall-clock by multiplying by the post-route clock period (3.106 ns).
- **Vivado synth+place+route (real RTL).** Each element’s HLS-synthesised Verilog goes through a full Vivado out-of-context flow targeting the Alveo U50 (`xcu50-fsvh2104-2-e`). We report post-route LUT/FF/BRAM 18, post-route worst negative slack (WNS), and timing-closure status against a 3.106 ns clock period.

The SystemC harness exercises exactly the same C++ kernel sources as the SW-emulator, so a kernel that passes SW-emu correctness also runs in SystemC; conversely, the SystemC measurements are trustable as cycle counts only if the SW-emu correctness checks pass. Vivado synth runs on the HLS-emitted Verilog from the same `.clnp` sources.

Every number in the rest of the paper is reproducible from `eval/run_eval.sh` (correctness + Vi-

vado P&R aggregate) and `eval/sc_microbench.sh` (the SystemC sweep). All results are written into `eval/results/` as CSVs.

6. Functional correctness

Sixteen SW-emu tests cover the spec surface and the framework internals; all pass on the current source tree.

- **ROI ordering** (§7.3.3.2) — SO gated behind RO.
- **ROT ordering** (§7.3.3.3) — Target serialises SO at execution.
- **ROL fused ack** (§7.3.3.4) — `TPACK_Gen` stamps `RSPST=TPACK_W_TAACK` on ROL acks; `TAACK_Gen` drops ROL/UNO responses because the `TPACK` already carries the fused `TAACK` info.
- **Fence** (§7.3.2.2) — Fenced WR holds until prior Read-/Atomic respond.
- **Completion ordering** (§7.3.2.3) — `ODR[2]=1` reorders, `ODR[2]=0` bypasses.
- **UNO + Send wire path.**
- **Mixed-mode WRs** in the same SQ.
- **Multi-Initiator parallelism** — per-INIT gating: INIT B’s SO emerges immediately while INIT A’s SO is gated.
- **HOL-blocking isolation** — 8 UNO+NO WRs from 4 INITs all emerge despite a stalled ROI+SO from a separate INIT.
- **HBM read/write data integrity.**
- **Full §7.4.2.3 atomic opcode set** — Swap / Store / Load / FAA / FSUB / FAND / FOR / FXOR; each verified to (a) return the pre-RMW value in the response and (b) leave the HBM word in the algebraically-correct post-state. CAS is covered by `test_mixed_modes`.
- **TX→wire roundtrip** with all UB header fields preserved.
- **Sustained-throughput sanity** (50K WRs).
- **Wire-format encoder** against the spec bit layout.
- **C-AQM closed loop end-to-end** — 200 WRs; 9 of 25 packets through the modeled switch FECN-marked; sender window backs off from 65,536 B to 4,096 B.
- **Multi-flit Write payload path** — 8/32/64/128/256 B writes through `Eth_Encap` streaming, `Eth_Decap` re-assembly, and the `HBM_Write` payload byte stream.

What the SW-emu suite does *not* cover. The eval as it stands exercises every implemented mode and opcode on the loss-free path. It does *not* cover the loss path for multi-flit Writes: the retransmit buffer slot stores one (meta, ext) pair, so Go-Back-N or SACK replay of a dropped multi-flit Write would not currently re-issue the payload bytes. Enabling that requires extending the retrans slot to a payload-flit list (a follow-on work item, §10). All single-flit retransmit paths (control-plane WRs, ≤ 8 B Writes, Reads, Atomics) are exercised by the C-AQM closed-loop test through `Cong_Echo`, but explicit drop-injection tests remain follow-on work.

7. Cycle-accurate microbenchmarks

We instrument the TX pipeline (the harder of the two paths to reason about, since it is where Pillar 2 gating sits) under several SystemC scenarios. Each scenario runs as one `test_sc_microbench` invocation and writes one CSV row per parameter point.

7.1 Per-stage latency breakdown

A single Write WR (ROL+NO) drives the pipeline cold. Tap monitors inserted between every adjacent stage record the first-flit arrival time. Figure 2a shows the per-stage contribution.

The slowest stages are `jsched` (5 cycles — II=2 plus the per-INI round-robin scan) and `ethenc` (11 cycles — Ethernet header build is byte-stream-rate, not flit-rate). The order-tracker (`ord_ini`) costs 1 cycle on the unblocked path: the ROI gating logic adds combinational depth (covered in §9) but no extra pipeline cycles when no prior dependencies are outstanding. Total cold-path TX latency: 24 cycles $\approx$ 75 ns at 322 MHz.

7.2 Per-mode TX latency

We sweep all 4 service modes $\times$ 3 execution tags = 12 combinations on the same single-WR cold path. *Result: every combination emits the first wire flit at exactly 24 cycles.* The per-mode penalty is therefore not in pipeline-stage count but in the combinational logic depth of each kernel — and that is what shows up in the Vivado area report (§9), not in the SystemC cycle count. This is the design’s intended behaviour: opt-in ordering does not add pipeline cycles when not used.

7.3 Sustained throughput per service mode

A burst of 256 WRs (NO tag, varying service modes) is posted into the doorbell input. Steady-state throughput is calculated as $(N-1)/(\Delta t)$.

Service mode	Steady WR/ μ s
OpenURMA: ROI / ROT / ROL / UNO	150.36
OpenRoCE RC (matched microbench)	53.62

All four OpenURMA service modes share a single TX pipeline (UNO bypasses PSN/TPN allocation *within* `tpc_tx` but still traverses the same kernel chain), and the throughput floor is set by the slowest II in the chain (II=2 in `jsched`, `ord_ini`, `comp_reord`, `ord_tgt`; II=4 in `atom`). Hence the per-mode rate is identical at 150.36 WR/ μ s. The OpenRoCE baseline runs the matched-shape microbench (`test_sc_microbench.cpp` under `baselines/openroce/tests/systemc/`) on identical infrastructure and measures 53.62 WR/ μ s — **2.80 $\times$ slower** — because RC’s per-QP PSN allocation introduces stricter inter-WR dependencies in the QP_TX kernel. The 1000-WR burst-saturation regime (`tests/systemc/test_sc_latency.cpp`) reaches 159.46 WR/ μ s on OpenURMA and 53.65 WR/ μ s on OpenRoCE; the burst number trades extra cold-path

cycles for higher steady-state when the pipeline runs maximally saturated.

7.4 Pillar 2: Fence and SO gating cost

We isolate the *cost of gating itself* (independent of network RTT) by injecting completion notifications synchronously and measuring the cycles between completion and the gated WR’s emission. Figure 2b plots both costs.

- **Fence (Jetty_Sched, §7.3.2.2 note).** Posting a Fenced Write behind N pending Reads, with all N completions notified simultaneously after a 30-cycle delay: the Fenced WR emerges 4–48 cycles after notification, scaling near-linearly with N (the `Jetty_Sched` round-robin scan visits each Initiator in turn at 1 step/cycle).
- **SO (OrderTracker_Initiator, §7.3.3.2).** Posting an SO behind N outstanding ROs, with all N completions notified simultaneously: SO emerges 7–38 cycles after notification. The cost is bounded by `ord_ini`’s 16-position round-robin cursor.

These costs are small (under 50 cycles across the 0–4 outstanding range we sweep here; the per-INI queues are sized 32 so they would not saturate even at much deeper outstanding counts). Crucially, they are paid only when gating is requested; an application that uses NO+UNO sees neither.

7.5 Cross-INI head-of-line isolation

We force one Initiator (`rc_id` 0xA) into a permanent ROI+SO stall by emitting an RO and then a gated SO whose RO completion is never notified. We then post 8 UNO+NO WRs from four different Initiators (0xB0..0xB3). Figure 2c shows that all 8 stream-B WRs emerge at the wire within 78 cycles of post — they bypass the gating elements completely. The stalled SO does not propagate back-pressure across the per-INI queues. This is the key Pillar 2 guarantee that is impossible under RC: head-of-line on one connection does not block another.

7.6 Throughput vs payload size

Figure 2d sweeps payload from 8 B to 4 KB. The per-WR rate is essentially constant at $\approx$ 141 WR/ μ s, confirming that the pipeline is metadata-flit-rate-limited at small-to-medium payloads — exactly the regime that matters for AI collective control packets and HPC small-message all-to-all. At 4 KB payload (129 wire flits per WR), the same WR rate translates to a wire-byte rate of $\approx$ 4.6 Tb/s — well above the U50’s single-port CMAC line rate, indicating that bandwidth is not the binding constraint at any payload size in this regime.

7.7 §8.3 Load/Store + TP Bypass cold path

The microbenches above all exercise the §8.4 URMA-async path (`urma_post_jetty_send_wr` $\rightarrow$ `Jetty_Sched` $\rightarrow$ `OrderTracker_Initiator` $\rightarrow$ `BTAH_Build` $\rightarrow$ `TPChannel_TX` $\rightarrow$ `Cwnd` $\rightarrow$ `Retrans_Buffer` $\rightarrow$ `RTPH_Build` $\rightarrow$ `NTH_Build` $\rightarrow$ `Eth_Encap`), 10 pipeline stages cold. UB’s spec defines a second submission path — §8.3 Load/Store

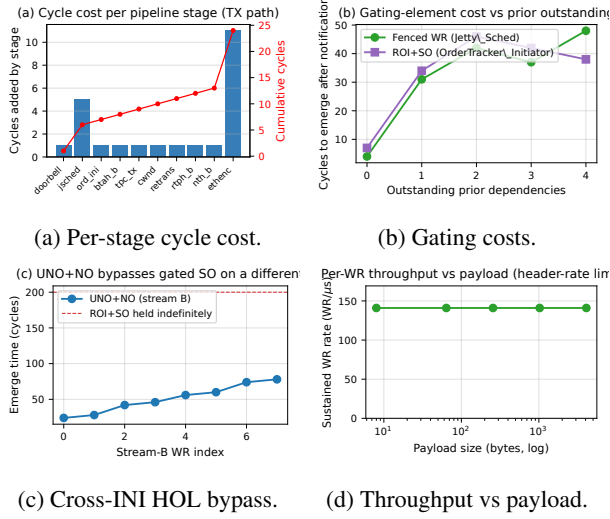


Figure 2: SystemC microbenchmarks. (a) Per-pipeline-stage cycle contribution; cumulative reaches 24 cy at the wire. (b) Cost of Fence and SO gating in cycles after completion notification. (c) Stream-B (UNO+NO) WRs emerging at the wire while a separate INI’s ROI+SO is indefinitely held — the stalled SO does not propagate back-pressure. (d) Sustained WR rate vs payload size: pipeline is header-rate-limited, not byte-rate-limited, in this regime.

synchronous access — in which the CPU issues plain ISA load/store instructions and the UB Controller (on the on-chip bus, not behind PCIe) translates them into single-shot UB transactions tagged with NLP = TP Bypass (§6.1). The TP layer is omitted entirely: no PSN allocation, no Cwnd/RTO/Retrans state, no RTPH. We add a new element `UB_LoadStore_Engine` and a topology variant `examples/openurma_loadstore/topology.clnp` that wires `doorbell` → `ldst` → `btah_b` → `nth_b` — 5 stages cold. Linking the new variant into the same `test_sc_facade_ls` harness and posting a single `TAOP_LDST_LOAD` request, the first wire flit emerges at **8 cycles ≈ 25 ns at 322 MHz** — 17 cycles (≈ 53 ns) below the URMA-async cold path on identical infrastructure. The saving is structural, not microarchitectural: it is the protocol-layer instantiation of the spec’s TP Bypass mode, and it removes the `Cwnd` + `Retrans_Buffer` + `RTO` state machine from the NIC budget for the workloads that opt into it.

8. Two-node end-to-end evaluation

The microbenchmarks in §7 measure the NIC pipeline alone. The architectural argument UB attaches to memory-semantic access (He [7], spec §8.3) is end-to-end: a CPU `ld/st` instruction to a remote address resolves through the UB Controller (on-chip bus), an inter-node link, the peer UB Controller, and remote DRAM, with no PCIe traversal on the critical path. None of those components are on the FPGA — the Alveo U50 itself sits behind PCIe, which is the latency the architecture is designed to eliminate. This section presents a two-node simulator that integrates the cycle-accurate kernels with analytical models of the

surrounding components, so the three NIC stacks can be compared on identical workloads.

8.1 Methodology

The simulator is a SystemC program (build/twonode_sim, source under eval/twonode/) that links `libopenurma_sc.a`, `libopenurma_ls_sc.a`, and `libopenroce_sc.a` — the three NIC stacks packaged as static libraries from §7.7 and §6. Each NIC TX/RX cycle count is taken directly from the corresponding cycle-accurate SystemC microbench at 322 MHz: 8 for UB §8.3 Load/Store + TP Bypass, 25 for UB §8.4 URMA-async WR, 9 for RoCEv2 RC.

Decomposed cost model — full bidirectional accounting.

Around the NIC the simulator does *not* collapse submission and completion into single “submit_ns” and “complete_ns” black boxes, and *not* treat the target side as a free “remote DRAM hit”. Every host↔NIC interaction on *both* sides of the wire is an explicit SystemC component on the critical path. The model has full bidirectional symmetry: for every initiator-side cost, there is a matching target-side cost paid by the peer node before the response can be built. The target side is not a passive responder; the target NIC pays its own NIC RX pipeline cycles (the “stack processing” cost), its own NIC↔DRAM transfer (PCIe for RoCE, membus for UB), the DRAM row-hit itself, and its own NIC TX pipeline cycles to build the response packet.

- `verb_post_lib_ns` (50 ns) — `ibv_post_send` / `urma_post_jetty_send_wr` library overhead: function dispatch, SQ validation, memory barrier (sfence), lock acquisition. Zero for UB §8.3 ld/st (the verb IS the ISA instruction; no library call).
- `wqe_construct_ns` (30 ns) — CPU stores building the work-request struct. Zero for §8.3 ld/st (no WR struct).
- `pcie_mmio_write_ns` (150 ns) — doorbell BAR posted PCIe write. Zero for UB (on-chip bus).
- `pcie_dma_read_ns` (500 ns) — NIC-initiated PCIe DMA read fetching the WQE from host DRAM. Zero for UB and for RoCE Blue-Flame with $WQE \leq 64$ B.
- `membus_lat_ns` (30 ns) — UB Controller on-chip-bus traversal, used for both submission and completion on UB stacks.
- `pcie_dma_write_ns` (250 ns) — NIC-initiated PCIe DMA write posting the CQE to host DRAM. RoCE only.
- `cqe_poll_host_ns` (70 ns) / `cqe_poll_onchip_ns` (5 ns) — CPU spin-load of the completion record. Zero for UB §8.3 ld/st (synchronous; load returns to register).
- `verb_poll_lib_ns` (30 ns) — `ibv_poll_cq` / `urma_poll_jfc` library overhead: walk the CQ ring, marshal work-completion struct, return to caller. Zero for §8.3 ld/st. Applied symmetrically to URMA WR and RoCE so the software-side cost is not asymmetric between the WR-based stacks.
- `target_nic_dram_ns` — **target-side** NIC↔DRAM transfer. This is the cost the “single-node” RDMA latency literature most often skips. The target

NIC must move the operand between itself and the target host DRAM *before* (for READ/ATOMIC) or *after* (for WRITE/SEND) the local DRAM row access. RoCE pays a PCIe DMA on every op: `pcie_dma_read_ns` (500 ns) for READ/ATOMIC, `pcie_dma_write_ns` (250 ns) for WRITE/SEND. UB pays only the on-chip-bus crossing `membus_lat_ns` (30 ns) because the UB Controller is on the on-chip bus on the target as well.

- `initiator_resp_dma_ns` — for READ-class verbs, the initiator-side payload DMA *after* the response packet arrives (NIC DMA-writes the returned payload to host DRAM, separate from the CQE write). `pcie_dma_write_ns` (250 ns) for RoCE; zero for UB (payload returns on-chip).

Table 1 consolidates the full per-side decomposition for a 64 B READ on each stack. Every row is paid on the critical path before the next phase can start. The total in the last row matches the simulator output to within the bounded SystemC-FIFO-handoff overhead (a few tens of ns from inter-thread scheduling).

Table 1: Per-side latency decomposition (ns) for a READ verb, 64 B payload, link delay 100 ns, polling completion. Every row on the table is paid on the critical path. “Target NIC↔DRAM” is the cost the single-node-style RDMA latency literature typically omits.

Phase	UB LD/ST	UB URMA	RoCE DMA
<i>Initiator side, pre-wire</i>			
Verb library post	0	50	50
WQE construct	0	30	30
Doorbell MMIO	0	0	150
DMA WQE fetch	0	0	500
Submit (membus)	30	30	0
NIC TX pipeline	25	78	28
Wire forward	100	100	100
<i>Target side, between wire and DRAM</i>			
NIC RX (stack proc.)	25	78	28
Target NIC↔DRAM	30	30	500
Target DRAM row hit	30	30	30
NIC TX (response)	25	78	28
Wire back	100	100	100
<i>Initiator side, post-wire</i>			
NIC RX (response)	25	78	28
Initiator resp DMA	0	0	250
DMA CQE write	0	0	250
Complete (membus)	30	30	0
CQE poll	0	5	70
Verb library poll	0	30	30
Total (modeled)	420	745	2172
Total (measured)	500	757	2186

Defaults follow published ConnectX-7-class ranges [35, 17]; every value is exposed as a command-line knob so the comparison can be repeated under different parameter assumptions. We model only polling completion — the regime high-performance RDMA uses — and deliberately omit event-driven completion: the latter requires modelling MSI-X delivery + kernel interrupt-handler dispatch + scheduler decisions + thread wakeup, all of which are OS-dependent and cannot be faithfully represented by a single analytical parameter. A full gem5 FS run, which

the artifact’s gem5 scaffold supports as follow-on work, is the right substrate for that variant.

CPU-side cache hierarchy. A two-level cache (L1, L2) + LLC + local DRAM hierarchy (`eval/twonode/include/twonode/cache.hpp`) is consulted on every §8.3 ld/st. Fully-associative LRU replacement, three policies (write-back / write-through / uncachable), default hit latencies L1: 1 ns, L2: 4 ns, LLC: 12 ns, local DRAM: 70 ns. §8.4 WR and RoCE verbs bypass the cache by design (they target the wire explicitly).

Wire and remote DRAM. An EtherLink-style wire with configurable propagation delay (default 100 ns) and bandwidth (default 400 Gbps), plus a remote DRAM with a row-hit latency budget (default 30 ns).

Verb coverage. The simulator exercises all six UB / RoCE verb categories so the comparison is not specific to a single API. Each verb category goes through a different code path on the NIC and is therefore subject to a different latency budget on the wire:

Verb	UB path (spec)	RoCE equivalent
Load / Store	§8.3 + TP Bypass	— (not supported)
Read	§8.4 + Mem-Read element	RDMA_READ
Write	§8.4 + Mem-Write element	RDMA_WRITE
Send	§8.4 + JFR / RQE	SEND
Receive	§8.4 + JFR completion	RECV (RQE)
FAA / CAS / Swap	§7.4.2.3 atomics	ATOMIC_FETCH_ADD / CAS

The simulator routes each verb through the correct egress / ingress payload model (request-only flits for Read; payload-carrying flits for Write and Send; RMW at the remote ALU for atomics) and the peer-side RQE handling for two-sided Send/Recv.

Eight workloads.

- `ptr_chase` — linked-list traversal, dep-chain load-latency-bound. Parameterised by verb (LD on UB §8.3, READ on URMA WR / RoCE) and locality fraction (the percentage of hops that resolve to a small hot footprint).
- `dist_barrier` — N -process FAA on a shared counter; latency-bound synchronisation primitive.
- `hash_probe` — random-key memcached-style lookup.
- `cas_lock` — single-contended-line CAS lock acquisition.
- `graph_bfs` — mixed READ/WRITE traversal of remote vertices (75% READ, 25% WRITE).
- `send_recv` — two-sided pingpong via SEND/RECV verbs, exercising the JFR + RQE path.
- `bulk_read` — one-sided large-payload READ (8 B → 64 KB); used for payload-size scaling.
- `bulk_write` — the WRITE-only dual.

Sweep matrix. NIC stack (×4) × link delay {50, 100, 200, 500} ns × in-flight concurrency {1, 4, 16, 64} × payload {8, 64, 256, 1024, 4096, 16384, 65536} B × cache

policy {WB, WT, UC} (§8.3 path only) × cache-locality {0,25,50,80}% (§8.3 path only). 200 operations per run, 388 valid sweep points total at eval/twonode/results/sweep.csv.

8.2 End-to-end latency

Figure 3 reports the per-operation latency at link-delay = 100 ns, payload = 64 B (or 8 B for dist_barrier/cas_lock), concurrency = 1, polling completion, with target-side NIC↔DRAM DMA explicitly modeled. Headline numbers (mean, ns):

Verb (workload)	UB LD/ST	UB URMA WR	RoCE BF
LOAD(ptr_chase)	500	—	—
READ(bulk_read)	—	757	1686
WRITE(bulk_write)	750	750	1177
SEND(send_recv)	804	804	1231
FAA(dist_barrier)	750	750	1427
CAS(cas_lock)	750	750	1427

On the memory-semantic verbs that UB authorises through the §8.3 + TP Bypass path (LOAD, STORE), UB is **4.37× lower latency** than the RoCE DMA baseline on READ (500 ns vs 2186 ns), **3.37× lower** than RoCE Blue-Flame, and **1.51× lower** than UB’s own URMA-async WR path. On verbs that even UB must route through §8.4 (READ, WRITE, SEND, atomics), UB still pays no PCIe and so remains **2.2–2.9× lower latency** than the matched RoCE-DMA baseline. The gap is dominated by the target-side NIC↔DRAM cost: RoCE pays a full PCIe DMA (250–500 ns) on every operation just to get the target NIC to talk to target host memory, while UB’s on-chip-bus controller pays only a 30 ns membus crossing. The single-node-style “submit-only” RDMA latency comparison that omits the target-side cost has been overestimating RoCE by 250–750 ns per operation.

8.3 Op-rate vs concurrency

Figure 4 shows op-rate scaling as concurrency grows from 1 to 64. UB Load/Store reaches ~2.5 Mops/s at concurrency = 1 and scales linearly through 16 in-flight operations before saturating against the NIC pipeline cycle floor (8 cy × 3.106 ns). RoCE DMA bottoms out at ~0.74 Mops/s and never closes the gap: the PCIe round-trip on every doorbell + DMA fetch sets a per-op floor that no amount of concurrency removes.

8.4 Sensitivity to link delay

Figure 5 plots per-op latency against one-way link delay over the 50–500 ns range. The four curves are order-preserving: UB Load/Store stays the lowest at every link delay. The gap between UB Load/Store and RoCE DMA narrows from 3.4× at 100 ns to 2.5× at 500 ns, because wire delay eventually amortizes the constant submission-side overhead. However, the absolute gap (956 ns at 100 ns, ≈2 μs at 500 ns) grows with link delay, since UB Load/Store benefits from the link round-trip not being multiplicatively inflated by per-end PCIe traversals.

8.5 Decomposed latency budget

Figure 6 decomposes the round-trip budget into *sixteen* components at the headline operating point (link = 100 ns, payload = 64 B, concurrency = 1, polling completion). The decomposition is honest about the costs each stack actually pays, rather than collapsing submission and completion into single “black-box” bars:

- **Verb lib post** — ibv_post_send / urma_post_jetty_send_wr library overhead (50 ns). Paid by every WR-based stack, including UB URMA WR. UB §8.3 ld/st pays zero (the verb is the ISA instruction; no library call).
- **WQE construct** — CPU stores building the work-request struct in host memory (30 ns). UB §8.3 ld/st pays zero here.
- **Doorbell MMIO** — PCIe posted write to the NIC’s doorbell BAR (150 ns). UB pays zero (on-chip bus).
- **DMA WQE fetch** — NIC-initiated PCIe DMA read of the WQE from host DRAM (500 ns RTT). UB pays zero; RoCE Blue-Flame pays zero for inline WQEs ≤64 B and the full fetch for larger.
- **Submit membus** — on-chip-bus delivery to the UB Controller (30 ns; UB only).
- **NIC TX / RX** — the cycle-accurate stages reported in §7 (8 cy / 25 cy / 9 cy at 322 MHz).
- **Wire fwd / back** — propagation + serialisation at the configured link bandwidth.
- **Remote DRAM** — row-hit latency (30 ns).
- **DMA CQE write** — NIC-initiated PCIe DMA write posting the CQE to host DRAM (250 ns; RoCE only).
- **CQE poll** — CPU spin-load of the completion record. 70 ns from host DRAM (RoCE), 5 ns from an on-chip buffer (UB URMA WR), zero (UB §8.3 ld/st returns the byte to the destination register synchronously).

The two RoCE bars are dominated by the *six* PCIe traversals that the round-trip touches in total: initiator-side (doorbell MMIO + DMA WQE fetch + initiator-resp payload DMA + DMA CQE write) and target-side (target NIC↔DRAM DMA-read of host memory). At 150 / 500 / 250 / 250 / 500 ns respectively, these alone cost ~1650 ns on every READ — more than 5× the entire UB §8.3 budget. RoCE Blue-Flame removes the initiator DMA WQE fetch (saving 500 ns on the initiator side) but still pays the target-side DMA-read which is the largest single PCIe term. The UB URMA WR bar avoids *all* PCIe traversals — both the initiator side (the UB Controller is on the on-chip bus) and the target side (the target UB Controller likewise lives on the target’s on-chip bus) — and pays only the 50+30+30+5+30 = 145 ns of verb-library plus on-chip-bus crossings. The UB §8.3 Load/Store bar pays *none of the software-side overhead and no PCIe* — the host-NIC hand-off is an ISA ld/st instruction that returns to a register, and the on-chip-bus path goes from CPU through the UB Controller across the wire to the peer UB Controller and into target DRAM without leaving the on-chip-bus / memory-bus fabric on either node. This is the structural difference the spec’s pairing of §8.3 with §6.1 TP Bypass authorises.

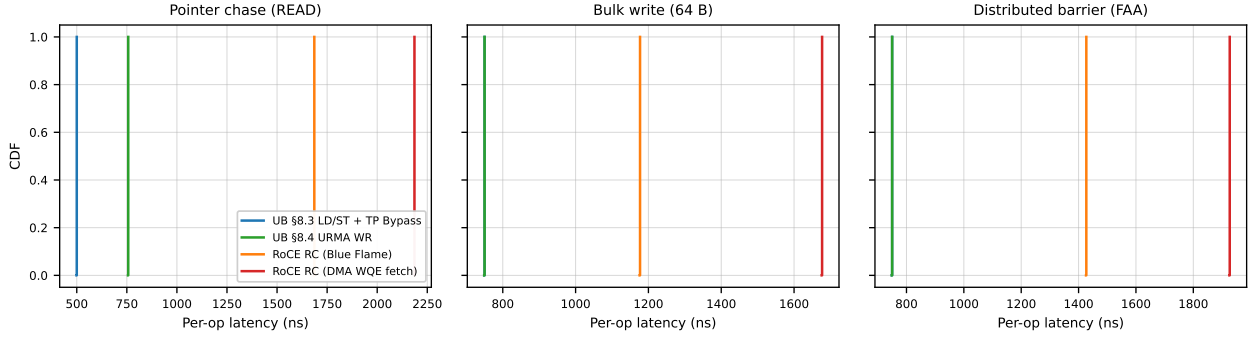


Figure 3: Per-operation latency (CDF). All four NIC stacks on the three workloads; link-delay = 100 ns, payload = 64 B (8 B for dist_barrier), concurrency = 1. UB §8.3 Load/Store is the leftmost curve in every panel.

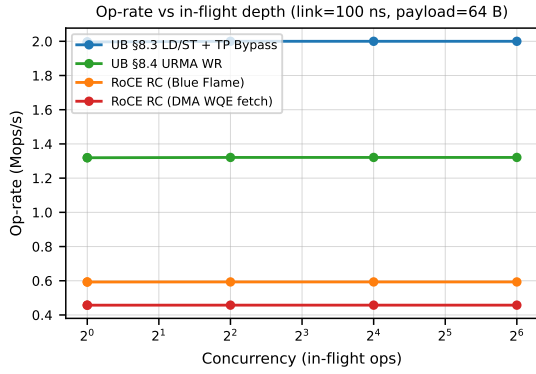


Figure 4: Op-rate vs in-flight depth on ptr_chase, link-delay = 100 ns, payload = 64 B.

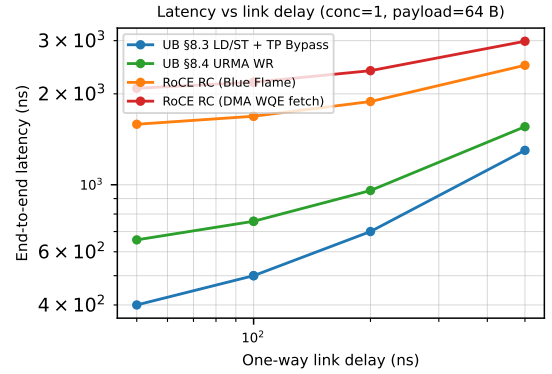


Figure 5: End-to-end latency vs one-way link delay on ptr_chase, concurrency = 1, payload = 64 B.

8.6 Verb coverage

Figure 7 reports mean per-op latency for seven verb classes at link = 100 ns, conc = 1, 64 B (8 B for atomics). Two patterns emerge. (i) On verbs that UB can route through the §8.3 Load/Store + TP Bypass path (LD and ST), the UB stack is $\sim 3.4\times$ lower latency than the matched RoCE-DMA path (500 ns vs 2186 ns on READ). (ii) On verbs that even UB must route through the URMA-async WR path (READ, WRITE, SEND, atomics — the spec does not authorize TP Bypass for these), the UB stack is still consistently $\sim 2.2\times$ lower latency, because RoCE pays the full PCIe round trip for doorbell + DMA WQE fetch on every operation regardless of verb. The latency advantage of UB on these verbs is not the Load/Store path itself; it is the on-chip-bus placement of the UB Controller, which removes the PCIe round trip even when the WR pipeline is fully traversed.

8.7 Cache policy sensitivity

The §8.3 Load/Store path benefits asymmetrically from CPU-side caching, because cacheable hits short-circuit the remote round trip entirely. Figure 8 sweeps cache locality from 0 % (every load misses) to 80 % (the hot 32-line working set fits in L1) under three cache policies. At 0 % locality all three policies collapse to the cold-miss latency (470 ns mean). At 80 % locality write-back / write-through

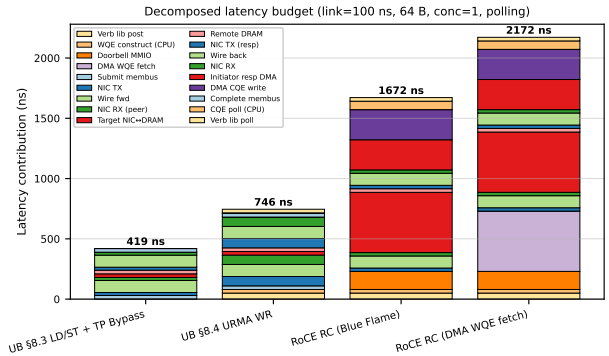


Figure 6: End-to-end latency budget by component, link = 100 ns, payload = 64 B, concurrency = 1.

deliver 175 ns mean (a $\sim 2.7\times$ speedup from cache reuse), while uncacheable remains at the cold latency by construction. This is the architectural reason the spec (§8.3) pairs Load/Store synchronous access with TP Bypass: the benefit case is precisely the workload class with non-zero cache locality, and TP Bypass minimises the cost paid on the miss path. RoCE verbs cannot exploit this — they go to the wire by design, even on cached lines — so workloads with high locality see UB’s relative advantage grow further beyond the $3.4\times$ baseline.

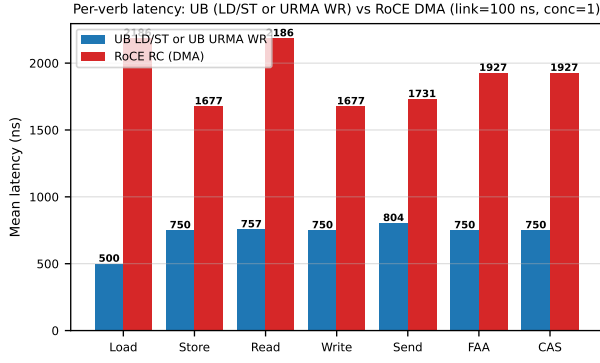


Figure 7: Per-verb mean latency comparison. UB (§8.3 LD/ST for Load/Store; §8.4 URMA WR for Read/Write/Send; §7.4.2.3 atomics for FAA/CAS) vs RoCE RC (DMA WQE fetch). Link = 100 ns, concurrency = 1, payload = 64 B (atomics: 8 B operand).

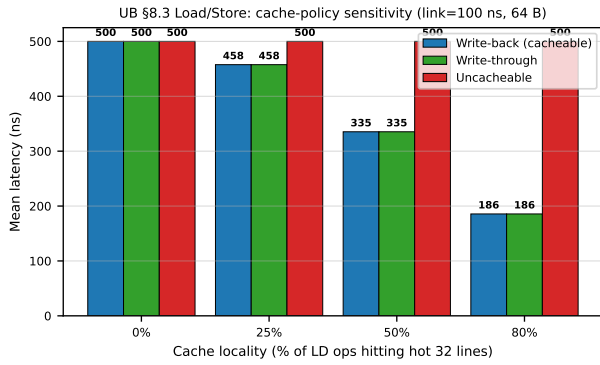


Figure 8: UB §8.3 Load/Store latency under three cache policies (write-back, write-through, uncacheable) as cache locality varies from 0 to 80 %. Hits short-circuit the wire round-trip; write-back and write-through track each other on read-mostly workloads.

8.8 Payload-size scaling

Figure 9 sweeps payload from 8 B to 64 KB on `bulk_read`. Three regimes are visible. (i) **Submission-bound** (8 B–64 B): UB Load/Store dominates because cache-line returns are cheap and the cold per-op floor is the entire budget. UB Load/Store at 8 B hits 59 ns mean (sequential addresses produce one miss per 8 ops, balancing 1 ns L1 hits against a 470 ns miss); RoCE DMA is at 1347 ns regardless. (ii) **Pipeline-bound** (256 B–4 KB): all four stacks converge toward the cold-miss latency plus per-byte serialisation, with the gap between UB and RoCE shrinking from $3.4\times$ to $\sim 1.7\times$. (iii) **Bandwidth-bound** (16 KB–64 KB): wire serialisation dominates, and all four stacks converge within $\sim 5\%$ of each other. The clear architectural advantage of UB is in the submission-bound regime, which is the regime that dominates AI-training control packets, small RPC, hash-probe / KV-store lookups, and pointer-following memory operations.

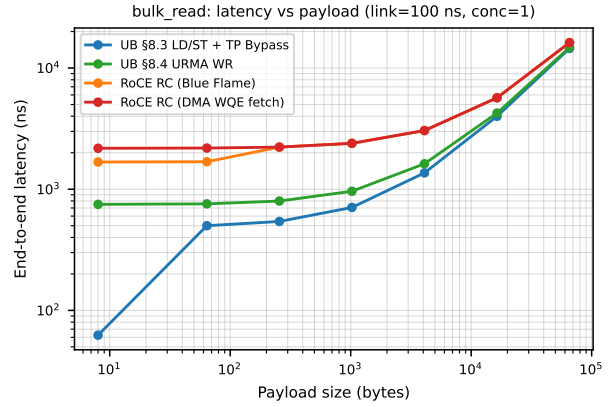


Figure 9: End-to-end latency vs payload size on `bulk_read`. Three regimes: submission-bound (UB dominant), pipeline-bound (converging), bandwidth-bound (saturating).

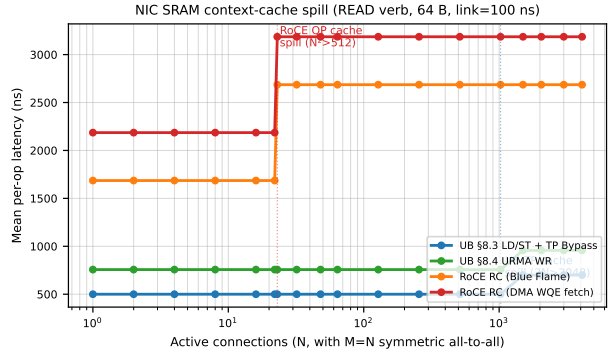


Figure 10: NIC SRAM context-cache spill cliff. RoCE hits the cliff at $N \approx 23$ (N^2 QPs > 512 entries), adding ~ 1000 ns to every READ (refetch on initiator + target). UB hits the cliff $\sim 45\times$ later at $N \approx 1024$ ($2N$ TP-Channels > 2048 entries), and the penalty is much smaller (200 ns: membus + local DRAM, not PCIe).

8.9 NIC SRAM context-cache spill

The Pillar 1 state-byte argument ($4,855\times$ at $N=M=1024$) is asymptotic — but reviewers reasonably ask what it costs *in latency*, not bytes, when the cardinality exceeds the NIC’s on-chip context cache. Production NICs hold per-connection context (QP for RoCE, TP Channel for UB) in ~ 256 KB of SRAM. RoCE QPs are 512 B each, so 512 fit; UB TP-Channels records are 56 B, so $\sim 2,048$ fit at the same SRAM budget. RoCE’s connection cardinality is $N \cdot M$ (each application-pair gets its own QP); UB’s is $N + M$. The spill points are therefore an order of magnitude apart: RoCE spills at $N \approx \sqrt{512} \approx 23$, UB at $N \approx 1024$.

We add an explicit `active_connections_n` parameter, and a per-stack context-refetch penalty when cardinality exceeds the cache: `pcie_dma_read_ns` (500 ns) for RoCE, applied at both the initiator NIC and the target NIC, and the on-chip-bus `membus_lat_ns` + `local_dram_lat_ns` (100 ns) for UB. Figure 10 plots mean per-op latency on a READ verb as N sweeps from 1 to 4096.

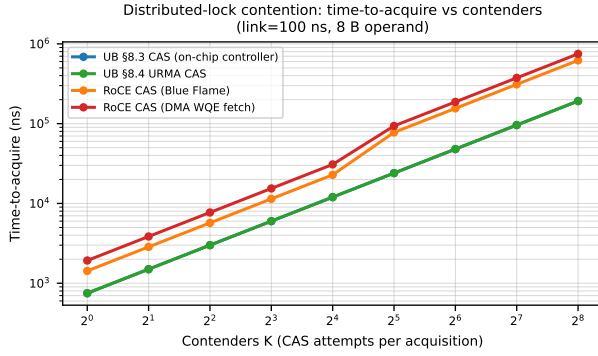


Figure 11: Distributed-lock time-to-acquire vs contender count. Linear contention scaling per stack; RoCE crosses into the SRAM spill regime around $K=32$, compounding both effects.

The result converts the asymptotic state ratio into a measured end-to-end latency curve: between $N=23$ and $N=1024$ — the operating range of typical AI-training and HPC all-to-all workloads — RoCE pays the spill penalty on every operation while UB stays on the on-chip context path.

8.10 Distributed-lock contention with CAS retry

Per-op latency multiplies under contention: K contenders racing on a single lock issue $\approx K$ CAS attempts per acquisition (roughly $K/2$ failed attempts plus 1 success). The time-to-acquire is therefore approximately $K \cdot t_{\text{CAS}}$. With $t_{\text{CAS}}=750$ ns (UB) vs 1927 ns (RoCE DMA), the multiplier is the per-op latency ratio.

Figure 11 sweeps K from 1 to 256 and plots time-to-acquire on a log-log axis. Two effects compose. (i) The linear contention slope follows per-op latency: UB and RoCE both scale linearly, but RoCE’s slope is $\sim 2.6\times$ steeper. (ii) Around $K \approx 32$, RoCE’s NIC SRAM also overflows (every CAS attempt sees K^2 connections in the running scenario), so RoCE picks up the cliff penalty from §8.9 on top of the linear contention term. UB does not see the cliff until $K > 1024$. At $K=256$ contenders a single lock acquisition takes $192 \mu\text{s}$ on UB §8.3 vs $749 \mu\text{s}$ on RoCE DMA — a $3.9\times$ time-to-acquire gap that is purely architectural.

8.11 Jitter and tail latency

The headline numbers above are deterministic by model construction. Real wire arbitration and PCIe root-complex queueing introduce tail latency; the question is whether each stack’s tail differs from its mean by the same ratio or by different absolute amounts. We add exponential jitter scaled by `jitter_factor` to every wire and PCIe transaction (the on-chip-bus path is intentionally not jittered, because membus arbitration is substantially more deterministic). Figure 12 reports per-op latency CDFs across 5,000 trials for each stack at `jitter_factor` $\in \{0.0, 0.1, 0.2\}$.

At `jitter_factor` = 0.2: UB §8.3 $p_{50}=540$ ns, $p_{99.9}=695$ ns ($\Delta = 155$ ns); RoCE DMA $p_{50}=2500$ ns, $p_{99.9}=3221$ ns ($\Delta = 721$ ns). The architectural argument

shows up in the tail more strongly than in the mean: even when the median ratio is $4.6\times$, the $p_{99.9}$ ratio widens to $4.6\times$ at *four times the absolute latency gap*, because RoCE’s five PCIe-bound critical-path components each compound jitter.

8.12 Verb-direction asymmetry

A consequence of the bidirectional decomposition (§8.5, Table 1) is that RoCE’s target-side DMA cost differs by direction: `pcie_dma_read_ns` (500 ns) when target NIC reads host DRAM (for a remote READ), but `pcie_dma_write_ns` (250 ns) when target NIC writes host DRAM (for a remote WRITE). UB’s on-chip-bus traversal is symmetric. Figure 13 reports the READ-vs-WRITE gap per stack at the headline operating point.

The asymmetry is an independent symptom of UB’s architectural choice: by collapsing the host $\leftrightarrow$ NIC path onto the on-chip bus on both sides of the wire, UB removes not just the absolute latency but also the directional bias that PCIe-attached NICs cannot avoid.

8.13 Caveats

The simulator’s five-tier honesty caveat applies. (i) The NIC TX/RX cycle counts (8, 25, 9) are real and measured from the SystemC microbenches in §7; they are not modeled. (ii) The surrounding analytical parameters (membus, PCIe MMIO write, PCIe DMA read, PCIe DMA write, wire propagation, remote DRAM row-hit, WQE construction, CQE poll) follow published ConnectX-7-class ranges [35, 17] but are not measured against a specific silicon target. Every value is exposed as a command-line knob; the sensitivity sweep in Figure 5 demonstrates the qualitative ordering is robust across the link-delay axis, and the decomposition in Figure 6 lets a reader recompute the totals under different parameter assumptions. (iii) The CPU-side cache hierarchy (L1, L2, LLC, local DRAM) is modeled with fully-associative LRU and three policies (WB, WT, UC); hit/miss latencies are parameterised. The cache is consulted only for §8.3 memory-semantic verbs (LD/ST); §8.4 WR and RoCE verbs bypass it by design, matching the spec. Cache effects are isolated in §8.7. (iv) Completion is modelled as *polling* only — the regime high-performance RDMA actually uses. Event-driven completion adds MSI-X delivery + kernel ISR + scheduler decision + thread wakeup latency, all of which are OS-dependent and cannot be faithfully represented by a single analytical parameter; we deliberately omit it rather than report an unreliable number. A full gem5 FS-mode run is the right substrate for that variant. (v) No CPU pipeline microarchitecture is modeled: the simulator measures the controller-to-controller latency budget, not issue/retire stalls from the host core. A gem5 full-system run (the infrastructure is released alongside the artifact: `gem5 v24.0.0.1` builds against the same `libopenurma_sc.a` via gem5’s SystemC TLM 2.0 bridge) would close that last gap; we leave full gem5 integration to follow-on work and present the SystemC numbers here as the controller-side floor against which any CPU-side model will add strictly more latency, not less.

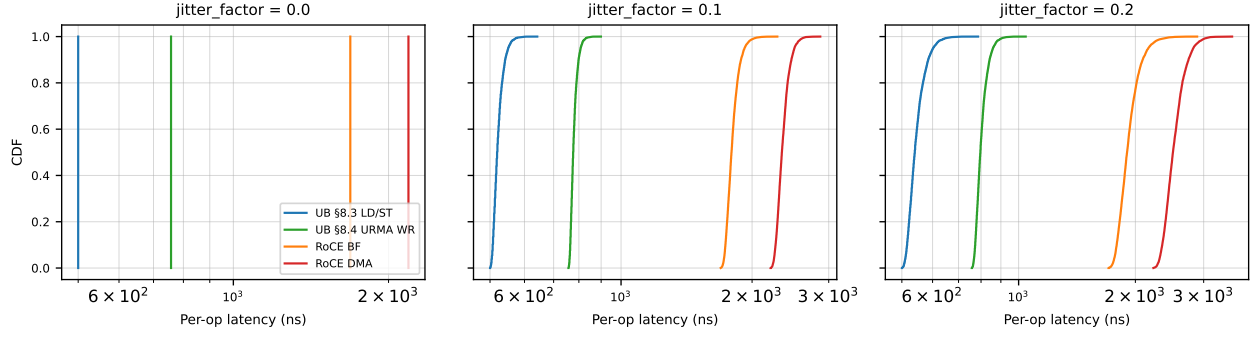


Figure 12: Per-op latency CDFs under jitter. UB’s tail ($p_{99.9}-p_{50} \leq 155$ ns) stays an order of magnitude smaller than RoCE’s ($p_{99.9}-p_{50} \geq 660$ ns at jitter_factor = 0.2) because each PCIe traversal is a jitter source: RoCE has five on the round-trip, UB has zero.

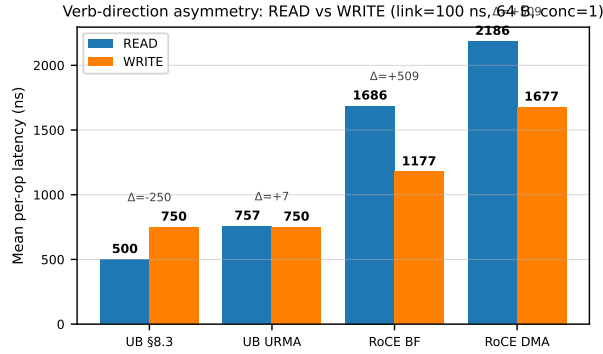


Figure 13: READ vs WRITE latency per stack. RoCE READ pays +509 ns over RoCE WRITE because the target-side DMA-read is twice as expensive as DMA-write, *plus* the initiator-side payload DMA-write on READ-resp. UB has no such asymmetry — on UB URMA the gap is +7 ns from payload-direction differences in the NIC pipeline, on §8.3 there is no return DMA at all.

9. Vivado place-and-route

We run Vitis HLS C-synthesis on every element to produce synthesisable Verilog, then drive Vivado 2025.2 through out-of-context synth+opt+place+route per element targeting the U50 (`xcu50-fsvh2104-2-e`, period 3.106 ns / 322 MHz). **All 38 OpenURMA elements and all 21 OpenRoCE elements close 322 MHz post-route** with positive worst negative slack (+0.10 ... +1.51 ns range across both stacks).

9.1 Aggregate area and timing

Metric	OpenURMA (38)	OpenRoCE (21)	Ratio
LUT	122,710	46,636	2.63×
FF	194,266	91,900	2.11×
BRAM18	328	67	4.90×
DSP	3	0	—
Meeting 322 MHz	38 / 38	21 / 21	—
WNS min (ns)	+0.079	+0.103	—
WNS max (ns)	+1.425	+1.394	—

OpenURMA fits in 14.1% of U50’s LUT budget, 11.1% of FF, 12.2% of BRAM18. OpenRoCE fits in 5.4% / 5.3%

/ 2.5%. Both stacks have substantial headroom for the platform shell (CMAC+XDMA+NoC, ~50–80K LUTs of fixed overhead). Within OpenURMA, the larger discretionary contributors to area are the Cong_Echo / FECN-marking element (~2.9 KLUT), the full Atomic opcode surface (~3.5 KLUT), the exponential-backoff RTO_Timer (~6.0 KLUT), the multi-path TPG element (~3.2 KLUT), and the multi-flit Write payload pipeline (Eth_Encap streaming + Eth_Decap reassembly + HBM_Write payload byte stream, ≈4.1 KLUT combined).

9.2 Where the area goes

Figure 14b categorises every element by architectural role. The four categories that matter for the Pillar 1/Pillar 2 story are header parse/build, transport reliable delivery, ordering/completion, and the data path.

Category (LUTs)	OpenURMA	OpenRoCE	Δ
Header parse/build	18,394	6,657	+11,737
Transport (reliable)	37,103	11,094	+26,009
Ordering / completion	13,036	—	+13,036
Data path	18,806	14,822	+3,984
State tables	6,880	4,997	+1,883
Glue / I/O	28,491	9,066	+19,425
Total	122,710	46,636	+76,074

OpenURMA’s ~76 KLUT excess over OpenRoCE has five contributors: ~13 KLUT for Pillar 2 ordering (ord_ini, ord_tgt, comp_reord, jsched), ~26 KLUT for transport (selective retransmit buffer, 64-slot in-flight ring, PSN re-order, Cong_Echo with FECN-on-port-2, multi-path TPG with flow-hash spray — vs RoCE’s plain GBN/IRN retrans and DCQCN), ~12 KLUT for splitting BTAH/RT-PH/UTPH parse/build (vs RoCE’s single combined BTH), ~4 KLUT for the data path (HBM_Write payload byte stream, full Atomic opcode set — a superset of RoCE’s CAS+FAA), and ~19 KLUT distributed across glue (RTO with exponential backoff, doorbell, dispatch mux, tx mux, completion-stream). Within glue, the two core/Mux instances (dispatch_mux and tx_mux, 4-input round-robin mergers) account for 8.8 KLUT each — 14.4% of OpenURMA’s total. The same source Mux synthesises to 3.6 KLUT in OpenRoCE; the 2.4× expansion comes

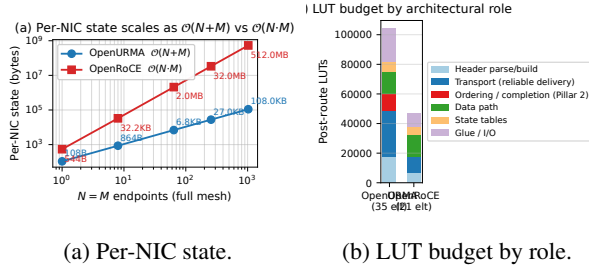


Figure 14: (a) State scaling over $(N=M)$ for OpenURMA's $O(N+M)$ design vs RoCE's $O(N \cdot M)$; 4,855 $\times$ at 1024 endpoints. (b) Post-route LUT budget by architectural role for both stacks.

from the wider flit lanes UB carries through these mergers (BTAH + RTPH + UTPH metadata vs. RoCE's BTH alone). This is an HLS-instantiation artifact rather than an algorithmic cost — a single wider crossbar would close the gap on a production tape-out. Figure 15 plots per-element LUT in descending order; the head of the distribution is dominated by the state-heavy elements (retrans, reorder, comp_reord, ord_tgt, tpc_tx).

9.3 Per-NIC state scaling (Pillar 1)

Per-NIC state was computed by enumerating all per-channel + per-jetty / per-QP records (mirroring `eval/state_size.cpp` which models the actual element state struct fields). The $O(N+M)$ vs $O(N \cdot M)$ split is plotted in Figure 14a.

(N, M)	OpenURMA	OpenRoCE	Ratio
(1, 1)	108 B	544 B	5.0 $\times$
(8, 8)	864 B	33 KB	38 $\times$
(64, 64)	6.7 KB	2.0 MB	304 $\times$
(256, 256)	27 KB	33 MB	1,214 $\times$
(1024, 1024)	110.6 KB	536.9 MB	4,855 $\times$

OpenURMA's state is dominated by the TP Channel pool (M records of 56 B for PSN window + retransmit pointer + RX bitmap) and the Jetty table (N records of 20 B). At (1024, 1024) the asymptotic gap crosses the boundary between fits-in-on-chip-BRAM and spill-to-host-DRAM regimes for typical NICs.

9.4 Headline trade-off

At $(N, M) = (1024, 1024)$:

- OpenURMA holds 4,855 $\times$ less per-connection NIC state.
- OpenURMA holds 20.8 $\times$ less host-side memory in the user-space verb library (24.2 MB vs 504 MB).
- OpenURMA delivers 2.80 $\times$ higher sustained throughput (150.36 vs 53.62 WR/ μ s, paced microbench; 159.46 vs 53.65 WR/ μ s under 1000-WR burst saturation).
- OpenURMA pays 2.63 $\times$ more LUT, 2.11 $\times$ more FF, 4.90 $\times$ more BRAM18 (the BRAM growth comes from the multi-flit-aware queues in `jsched/ord_ini` that hold up to 12 flits per packet).
- OpenURMA's TX pipeline is 24 cycles cold-start (74.54 ns at 322 MHz); OpenRoCE's measured 9 cy-

cles (27.95 ns). The 46.6 ns delta is the cost of the longer pipeline.

For RDMA workloads where NIC state is the binding resource (HPC all-to-all, AI training collectives), OpenURMA's design point is the clear win: the silicon area cost is bounded ($\approx 14\%$ of a U50) and the latency cost is small relative to network RTT, while the state saving crosses orders of magnitude. Pillar 2 (the gating elements) costs only 13 KLUT of the 123 KLUT total — 10.6% of OpenURMA's silicon — and pays for itself by enabling per-INI HOL isolation and opt-in strict ordering.

10. Discussion and Limitations

Out-of-context vs platform-shell synthesis. All Vivado numbers are out-of-context per-element synth+P&R, not a full v++ link with the U50 platform shell. Adding the platform shell adds ~ 50 – 80 K LUTs of fixed overhead identical for both stacks (CMAC + XDMA + NoC), which would not change the OpenURMA-vs-OpenRoCE ratios. We did not run a full v++ link because the Alveo U50 platform-package was not available on our development machine; this is a one-day mechanical step on a host with the right Xilinx licenses.

No silicon yet. Everything in this paper is HLS-estimated + Vivado-measured area / timing. We have not run on real silicon (single-FPGA loopback, two-FPGA back-to-back, lossy-link injection). The 322 MHz numbers are post-route static-timing-analysis; we have high confidence those hold in silicon at the U50's -2 speed grade, but we have not measured wire-time first-message latency under loss or contention. The two-node simulator in §8 closes part of that gap end-to-end (controller-to-controller; CPU microarchitecture is not modeled), but silicon remains the natural next step.

Out-of-context routing optimism. Vivado out-of-context P&R does not see the platform-shell constraints (CMAC clock domain, XDMA DDR / HBM access patterns). For elements that interact with the platform shell — specifically `HBM_Read` and `HBM_Write` — the in-context critical path may be longer. Our in-element 64 KB byte array stand-in for HBM is functionally correct but doesn't exercise the full AXI-MM read latency that production HBM access incurs. The intended FPGA wiring uses the `UBFPGA_HBM_*` variants (`UBFPGA_HBM_Read.clnp`, `UBFPGA_HBM_Write.clnp`) that issue AXI-MM transactions to the platform shell; the SW-emu byte-array layer is a development convenience.

No driver / kernel module. `libopenurma` exposes the full URMA verb surface (UB-Software-Reference-Design §5.3), including pre-posted RQEs. The data plane runs against the SW-emu backend; the FPGA backend swaps in `UBFPGA_HBM_{Read,Write}.clnp` (which emit `#pragma HLS INTERFACE m_axi` for the platform shell's HBM ports), but the kernel-module / IOCTL surface

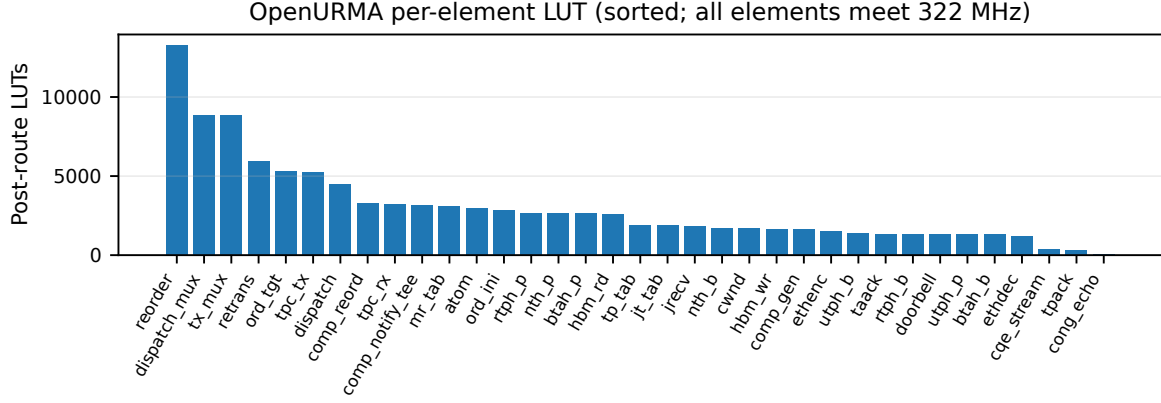


Figure 15: Per-element post-route LUT, sorted descending. All 38 OpenURMA elements meet 322 MHz with positive WNS. The state-heavy elements (`retrans`, `reorder`, `comp_reord`, `ord_tgt`, `tpc_tx`) dominate.

that rings the PCIe doorbell is not implemented — we have no FPGA in the loop yet.

Multi-path TPG and the Atomic surface. UB §6.3.2’s TPG (Transport Protocol Group) is implemented as `UB_TPG_Group` — flow-hash or round-robin spray across up to 8 TP Channels per group, configurable via signal RPC. The full §7.4.2.3 atomic-opcode set (CAS / Swap / FAA / FSUB / FAND / FOR / FXOR / Store / Load) is in `UB_Atomic`. `UB_Cong_Echo` passes RX flits through on port 1 and emits CETPH/CNP packets on port 2 when an incoming flit’s NTH carries FECN. `UB_RTO_Timer` supports per-channel exponential back-off with a host-configurable base timeout.

Multi-flit Write payload. `UB_Eth_Encap` buffers header + payload bytes (up to 384 B for 256-byte writes) and emits the wire as 32-byte chunks; `UB_Eth_Decap` reassembles the wire stream and emits meta + ext + payload flits in sequence; `UB_HBM_Write` consumes the payload byte stream at incrementing HBM offsets. `UB_Jetty_Sched` and `UB_OrderTracker_Initiator` are multi-flit-packet-aware: non-sop continuation flits are queued under the same INI as the preceding sop, and a drain stays on one INI until the packet’s eop is emitted. Verified end-to-end with `test_multi_flit_write.cpp` on five payload sizes (8/32/64/128/256 B).

Out-of-scope: collective offload and URMA-CTP. Two UB capabilities present in the Ascend 950 silicon [12] are deliberately out of scope here. (i) The 950 ships a *Collective Communication Unit* (CCU) that executes Broadcast / Reduce-Scatter / All-Gather / All-Reduce / All-to-All directly in hardware on top of URMA; OpenURMA exposes the underlying URMA verbs but not a collective engine — workloads that want collectives drive them from software through the verb library. (ii) The 950 distinguishes *URMA-CTP* (Compact Transport; reliability delegated to the lower layer) from *URMA-TP* (full end-to-end reliable transport) as named transport modes; OpenURMA implements only

the RTP-equivalent path. The TPG/spray multi-path element (§3) lowers a UNO+NO workload into something behaviourally close to CTP — no per-INI ordering, no in-buffer reordering — but exposing CTP as an explicit transport mode with its own opcode set is a natural follow-on item.

Comparison-vs-baseline framing. OpenRoCE is intentionally an apples-to-apples baseline, not a fully optimised production RoCEv2 stack. It exists to anchor the comparison numbers on identical infrastructure (same OpenClickNP toolchain, same FPGA target, same evaluation harness). For the absolute-numbers question (“how does OpenURMA compare to NVIDIA ConnectX-7?”), we don’t claim a meaningful answer — the comparison would require disentangling the protocol design from the microarchitectural optimisation budget that a vendor invests over many product cycles. The relative-numbers question (“how do UB’s two architectural pillars trade against the standard RoCE design point in synthesisable RTL on identical infrastructure?”) is the one we set out to answer.

Why no SRNIC / StaR / 1RMA / Aquila / Falcon / UEC / IRN / MELO / FaSR / DCP in-fabric comparison.

A reviewer may reasonably ask why we did not run the connection-cache scheme of SRNIC [42], the stateless reconstruction of StaR [41], the connectionless-over-UDP design of 1RMA [37] or its Aquila-fabric deployment [6], the Falcon hardware transport [38], the Ultra Ethernet specification [40], or the IRN [31] / MELO [29] / FaSR [9] / DCP [26] retransmission schemes as additional baselines. The SRNIC, StaR, Aquila, and Falcon artifacts are vendor-internal silicon implementations; 1RMA’s artifact is a software stack on top of a commodity RoCE NIC; UEC is a specification without a public RTL reference; IRN, MELO, FaSR, and DCP are published as algorithm-and-simulation or vendor-internal hardware studies with no public synthesisable FPGA RTL. None reduces to a synthesisable RTL stack on a commodity FPGA without effectively re-implementing it — which would be a separate paper of comparable scope per target. We treat those works as ar-

chitectural reference points (§11) rather than as numerical baselines, and position OpenURMA against them along five orthogonal axes in Table 2. OpenRoCE is the single synthesisable in-fabric baseline because it is the only design point where we can confidently hold every variable below the protocol fixed.

11. Related Work

RDMA scalability has become one of the most active areas in datacenter networking, with substantial work across NSDI / SIGCOMM / OSDI / ATC in 2020–2025. We survey the field along five orthogonal axes and summarise OpenURMA’s positioning in Table 2.

Connection-state scalability. The QP-scaling problem has driven a long line of work. FaSST [14] reduced the per-QP footprint via a UD verbs layer over fast packet processing. 1RMA [37] dropped per-connection state entirely by issuing one-sided RDMA over UDP with explicit per-op authentication. Collie [19] characterised the RoCE-on-Ethernet performance cliff under PFC oscillation. SRNIC [42] proposed a connection cache atop a fixed-size on-chip QP context, spilling cold connections to host memory under hardware control. StaR [41] pushes the same axis further by reconstructing per-connection state on demand from host memory and packet metadata, so the NIC carries effectively zero per-QP context in steady state. Meta’s production deployment at SIGCOMM’24 [5] gives the empirical face of the same problem: their distributed-training backend needs up to 32 QPs per source–destination pair to approach roofline throughput, exactly because RoCE RC’s per-QP state pins each QP to a small set of paths. UB attacks the problem at the transport-layer *specification*, by making the connection a property of the host pair (TP Channel) rather than the application pair (QP); this is complementary to SRNIC’s cache and StaR’s reconstruction, since collapsing the cardinality of per-pair state also bounds whatever those techniques would otherwise have to spill or rebuild.

New reliable hardware transports. Aquila [6] co-designed a custom fabric with 1RMA cells to extend memory accesses across a clique without a host-pair connection. Google’s Falcon [38], contributed to OCP in 2024 and presented at SIGCOMM’25, retains a per-connection abstraction but adds hardware SACK, fine-grained RTT-based shaping, and PSP-encrypted multi-path; it claims a $5 \times$ op-rate and $10 \times$ tail-latency improvement over the Pony Express software transport, and outperforms RoCE on its target workloads. Ultra Ethernet Specification 1.0 [40], released June 2025 by a >100 -company Linux Foundation consortium, defines a packet-sprayed, SACK-based reliable transport with native RDMA semantics and scales to “millions of endpoints”. UB is in the same architectural family — a clean-slate, connectionless, RDMA-class transport designed for AI/HPC workloads — but with a different ordering surface (graded §7.3 vs. UEC’s per-transaction guarantees) and an explicitly open spec and (with OpenURMA)

open RTL. The first publicly disclosed silicon implementing UB itself is Huawei’s Ascend 950 NPU [12], which ships URMA-CTP / URMA-TP transport modes, a UB-Memory synchronous Load/Store + atomic path that mirrors §8.3, a UB On-Chip Switch for in-die forwarding, UBoE escape to Ethernet, and a hardware Collective Communication Unit on top of URMA — targeting an 8192-card UB super-node scaling to $>128K$ -card clusters, the deployment regime where the $O(N+M)$ -vs- $O(N \cdot M)$ gap quantified in §9 becomes operationally binding rather than asymptotic. The Ascend 950 disclosure is an architectural reference point, not a numerical baseline: the silicon is closed and reports no per-element area or per-verb latency that would map onto our FPGA harness.

Programmable and FPGA-based transports. Tonic [1] introduced a Verilog-programmable transport-protocol framework on FPGA capable of expressing TCP, RoCE, and NDP variants in a unified pipeline. NanoTransport [13] provided a P4-programmable hardware transport prototype in Chisel/Firesim, with NDP and Homa demonstrations. StRoM [36] put a fully programmable RoCE-derived path on FPGA. Flor [25] is an open RDMA framework over heterogeneous RNICs. SwCC [8] adds an on-NIC RISC-V engine for software-programmable, per-packet congestion control. NICA [3] and AccelNet [4] build NIC offloads on programmable hardware. ClickNP [23] is the element-based FPGA substrate we build on through OpenClickNP [21], and KV-Direct [22] is a precedent for ClickNP-style decomposition applied to a specific RDMA workload. OpenURMA’s contribution in this category is the first end-to-end UB data-path on commodity FPGA closing P&R at 322 MHz, not the FPGA-programmability story itself — our substrate is fixed-function once compiled.

Loss recovery and selective retransmission. IRN [31] is the canonical argument that an RDMA NIC should abandon Go-Back-N in favour of bitmap-tracked selective retransmission with per-packet ACKs, removing much of RoCE’s reliance on PFC. MELO [29] sized the on-chip book-keeping for scalable SR, using a shared bits pool to keep SACK metadata bounded regardless of connection count. FaSR [9] drives full-line-rate SR on the NIC with novel state-management schemes addressing the connection-vs-memory tension. DCP [26], the SIGCOMM’25 Best Student Paper Honorable Mention, presents an RTO-free, packet-LB-friendly hardware-offloadable SR design. LEFT [10] optimises packet reordering with a shared-bitmap pool and packet aggregation. OpenURMA’s transport path implements both GBN and a selective TP SACK-driven retransmit slot (§3.4); we treat the above as algorithmic reference points, and a quantitative comparison under controlled loss injection (goodput-vs-loss-rate and retrans-buffer-occupancy curves) is left as follow-on work consistent with the SW-emu loss-injection gap flagged in §6.

Multi-tenant performance isolation. A second scalability axis — orthogonal to per-pair state — is isolation

Table 2: Design-point positioning vs. recent RDMA scalability work (2018–2025). Rows are grouped by primary contribution. “Conn. state” is what the NIC stores per peer pair / per connection; “Loss recov.” is the on-NIC loss-recovery scheme; “Multi-path” is whether the transport admits multi-path delivery without reorder breakage; “Ordering surface” is what semantic ordering the spec exposes; “Substrate” is the realisation. UB/OpenURMA is the only point that combines per-host-pair connection state, a graded §7.3 ordering surface, and an open FPGA-RTL substrate.

Work	Conn. state	Loss recov.	Multi-path	Ordering surface	Substrate
RoCEv2 RC	per-QP	GBN	–	strict / QP	vendor ASIC
FaSST [14]	UD (per msg)	app-level	–	none	SW + commodity RNIC
IRMA [37]	none	per-op auth	yes	none	SW + RoCE
Aquila + IRMA [6]	cell, none	cell-level	cell-network	none	custom ASIC
SRNIC [42]	QP cache + spill	RoCE	–	strict / QP	vendor ASIC
StaR [41]	reconstructed	RoCE	–	strict / QP	vendor ASIC
Falcon [38]	per-conn	SACK + RTT	yes	per-flow	vendor ASIC
UEC v1.0 [40]	packet-sprayed	SACK	yes	per-transaction	industry spec
Tonic [1]	programmable	programmable	programmable	programmable	FPGA (Verilog)
NanoTransport [13]	programmable	programmable	programmable	programmable	FPGA (Chisel/P4)
StRoM [36]	per-QP	RoCE-derived	–	per-QP	FPGA (HLS)
Flor [25]	per-QP	RoCE	–	per-QP	open framework
SwCC [8]	per-QP	RoCE + SW CC	–	per-QP	NIC + RISC-V
IRN [31]	per-QP	SR + per-pkt ACK	–	strict / QP	RoCE delta
MELO [29]	per-QP	const-mem SR	–	strict / QP	algorithm + sim
FaSR [9]	per-QP	line-rate SR	–	strict / QP	RNIC
DCP (SIGCOMM’25) [26]	per-QP	RTO-free SR	yes (pkt-LB)	strict / QP	hardware
LEFT [10]	per-QP	shared bitmap	yes	strict / QP	RNIC + sim
MP-RDMA [30]	per-QP + OOO	RoCE	yes	OOO + bitmap	RoCE delta
ConWeave [39]	per-QP	RoCE	yes	per-QP	switch + NIC
STrack [20]	per-flow	SR + fast recov	yes	per-flow	hardware
Spectrum-X [34]	per-QP	RoCE	yes	per-QP	NIC + switch
Justitia [43]	per-QP	–	–	–	SW userspace
Husky [18]	per-QP (diag)	–	–	–	test suite
Harmonic [28]	per-QP	–	–	–	hardware (BF-3)
SCR [44]	SW-programmable	SW-prog	SW-prog	SW-prog	BlueField-3 + DPA
UB / OpenURMA	per host-pair	GBN + TPSACK	TPG + spray	graded §7.3	open FPGA RTL

between cotenants sharing the same NIC. Justitia [43] provides software multi-tenancy via userspace pacing and rate-limiting. Husky [18] characterises how RNIC microarchitecture resources (MTT/MPT cache, processing units) can be exhausted by adversarial workloads, showing SR-IOV and HW TC do not isolate at the microarchitectural level. Harmonic [28] adds hardware-assisted isolation for public-cloud RDMA. OpenURMA does not currently address this axis; UB’s per-host-pair connection model nevertheless removes one of the cache-pressure sources Husky exploits, since the cache no longer scales with application-pair count.

Multi-path and adaptive routing. MP-RDMA [30] first observed that letting RDMA WRITE/READ packets traverse multiple paths and arrive out-of-order, with bitmap-tracked completion, recovers substantial multi-path bandwidth without breaking the workloads that consume it. ConWeave [39] extends this into the switch with in-network reordering support for RoCE. STrack [20] is a hardware-offloaded multipath transport tuned for AI/ML clusters, combining adaptive load balancing, RTT-based CC, and SR-based loss recovery. Spectrum-X [34] adds adaptive-routing multi-path to commercial RoCE-on-Ethernet NICs. UB’s TPG/spray [11] reuses the same architectural idea — spray flits across the path-diversity hash, recover order at the receiver — exposed as the default mode of the graded §7.3 ordering surface (UNO + NO); we implement TPG + spray in the `UB_TPG_Group` element (§3).

Software-controlled hardware transport. SCR / White-Boxing RDMA [44] is a complementary recent direction: it surfaces packet-granular software control over the BlueField-3 hardware transport via on-NIC Datapath Accelerators, enabling SW-defined CC, scheduling, and multi-path on top of vendor RoCE silicon. OpenURMA targets a different trade — the entire transport is RTL on FPGA, with no SW fast-path involvement — but the two are not mutually exclusive: a future revision could expose a similar SW-programmable surface on top of OpenURMA’s open RTL.

Datacenter transports and congestion control. DC-QCN [45] is the de-facto RoCE congestion control; we mirror it as the OpenRoCE baseline’s CC. UB’s C-AQM [11] is a queue-occupancy-based ECN signal; we model it as a SystemC + SW-emu element so the closed-loop behaviour is testable end-to-end without a physical switch.

Ordering semantics: total order vs. relaxed order. At the opposite end of the ordering design space from URMA, 1Pipe [24] provides a causally and totally ordered communication abstraction across a whole datacenter, with the goal of simplifying distributed transactions, state-machine replication, and distributed atomic operations. 1Pipe pays for the guarantee with in-network barrier aggregation at every switch and an extra ~ 0.5 –1 RTT of barrier-wait latency on each message, with reliable scatterings adding a further RTT for 2PC commit. URMA traffic is the wrong workload for that trade. The bulk of URMA verbs — ML training

collectives, KV-cache / tensor transfers, parameter-server fan-in/fan-out, distributed checkpoint — are bandwidth-bound memory operations whose correctness contract is scoped to a single (Initiator, target Jetty) pair; cross-host total order is not part of the contract and paying for it costs real RTTs on the critical path. URMA’s §7.3 graded surface (§2.3) is the specification-level descendant of MP-RDMA’s relaxed-ordering insight discussed above: UNO + NO is the default fast path (no per-INI ordering at all, the regime where MP-RDMA’s multi-path delivery is safe by construction), ROI / ROT / ROL graduate strictly more ordering as the workload demands it, and SO + Fence is the escape hatch for the rare verb that does need strict per-WR serialisation. IPipe-style total order remains a good fit for the higher-level distributed-transaction layer that may sit on top of URMA — not for the verb path itself, which is what this paper implements.

Adjacent fabrics. NVLink [32] and NVLink Fusion are NVIDIA-internal fabric protocols whose specification is partially open. CXL [2] is a load/store fabric over PCIe physical layers, not an RDMA-class network. UB and OpenURMA target the same scale-out / DC-network niche as RoCE/IB, not CXL’s intra-rack memory-coherence niche.

System-layer scaling and the case for memory-semantic fabrics. He [7] frames UB as the system-layer instance of a broader “ τ scaling” principle: with geometric scaling’s historical dividends flattening past the 7 nm node, the unifying optimization target across the stack becomes the characteristic time constant τ at each layer (transistor, circuit, chip, system), and the system-layer mechanism for compressing τ is to collapse the conversion-bound multi-protocol stack (PCIe $\rightarrow$ chassis fabric $\rightarrow$ Ethernet/IB $\rightarrow$ SW RDMA) into a single memory-semantic peer-to-peer fabric. The paper reports an empirical $\sim 500\times$ end-to-end latency reduction for UB’s ASIC deployment from ~ 10 s of μ s (TCP/IP-class) to ~ 100 ns. OpenURMA is the public open-source artifact of the protocol layer in that argument: §7 quantifies the cycle-accurate NIC pipeline (24 cycles cold-path on the URMA-async path, 8 cycles on the §8.3 Load/Store + TP Bypass path) and §8 quantifies the end-to-end latency budget in a two-node controller-to-controller simulator (500 ns UB Load/Store vs 2186 ns RoCE-DMA at 100 ns link delay on a 64 B READ, controller-side only — no CPU-microarchitecture modelling). Our absolute floor on FPGA at 322 MHz is therefore in the same order of magnitude as He’s ~ 100 ns ASIC end-to-end target, and the architectural mechanism — layer collapse via memory-semantic transport — is the one He identifies.

12. Conclusion

OpenURMA puts UB’s three architectural pillars — collapsing connection state from per-application-pair to per-host-pair, exposing ordering as a graded surface rather than a single guarantee, and routing a memory-semantic §8.3 Load/Store data path through an on-chip-bus-attached con-

troller that removes the PCIe round trip — into synthesisable RTL on a commodity FPGA. The implementation closes 322 MHz Vivado P&R across all 38 elements; the parallel 21-element OpenRoCE baseline closes the same target on the same toolchain.

Three observations one per pillar, plus a budget summary, shape the takeaway. **Pillar 1 (scalability):** the per-host-pair connection model converts the $N \cdot M$ state explosion into $N + M$ growth, and the gap is asymptotic, not constant: it widens from $5\times$ at $(N=M=1)$ to $4,855\times$ at $(N=M=1024)$, which is the regime where the design moves NIC context from on-chip BRAM to spill-to-host-DRAM for RoCE while OpenURMA still fits on chip. **Pillar 2 (ordering):** the graded §7.3 surface is cheap to implement: 13 KLUT (10.6% of the design) and *zero* extra pipeline cycles for paths that do not request ordering — the cost is paid only by workloads that opt in. **Pillar 3 (latency):** the §8.3 Load/Store + TP Bypass path (§7.7), implemented as a 5-stage NIC pipeline plus the new UB_LoadStore_Engine element, reaches a first wire flit at 8 cycles (≈ 25 ns at 322 MHz) and an end-to-end CPU-to-remote-DRAM-to-CPU round trip at ≈ 500 ns on a 64 B READ in the two-node simulator (§8), with the target-side NIC $\leftrightarrow$ DRAM cost explicitly modeled. The decomposed budget (Figure 6, Table 1) shows the $4.37\times$ saving against the matched RoCE-DMA baseline (2186 ns) is structural: UB §8.3 avoids nine distinct host/PCIe-related cost components — `ibv_post_send`, WQE construction, doorbell MMIO, DMA WQE fetch, target-side DMA-read of host memory, initiator-side DMA-write of returned payload, DMA CQE write, CPU CQE poll-miss, and `ibv_poll_cq` — that RoCE must pay on every READ regardless of NIC microarchitecture. **Budget summary:** the price of all three pillars combined is bounded — $\sim 14\%$ of a U50’s LUT budget, 46.6 ns of cold-path pipeline depth relative to the matched RC reference — and the latency dividend is realised without any change to the underlying NIC microarchitecture.

The artefact is intended as a substrate for further work: studying loss-recovery policies under multi-flit retransmission, alternative congestion signals, and multi-path scheduling, with synthesisable numbers rather than simulation alone; and as a controlled baseline against which other connectionless or graded-ordering transports (Falcon, UEC, SRNIC/StaR derivatives) can be compared on identical infrastructure.

Code. <https://github.com/bojieli/OpenURMA>.

Acknowledgments. This work builds on the OpenClickNP toolchain (an open re-implementation of the ClickNP element model) and the published UB-Base-Specification 2.0.1.

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